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Ernest J. Sternglass

Low-Level Radiation

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attention to radiation
damage to infants and the unborn.





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LOW-LEVEL RADIATION

Ernest J. Sternglass

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An Intext Publisher

For Marilyn and our children, Daniel and Susan

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SBN 345-02516-4-125

First Printing: March, 1972

Printed in the United States of America

Cover photograph by Ed Eng

BALLANTINE BOOKS, INC.
101 Fifth Avenue, New York, N.Y. 10003

Acknowledgments

This book could not have come into being without the understanding, concern and support of many individuals whose help I was privileged to receive during the past ten years in which the events recounted here took place.

Although it is impossible to list all those to whom I have become deeply indebted, there are a few individuals whose help went far beyond anything I shall ever be able to acknowledge adequately.

First among these is my wife Marilyn, who not only stood by my side throughout these years, but also provided the constant counsel, encouragement and understanding needed in the long and arduous task of writing this book.

And it was the great personal dedication of my editor, Joel Griffiths, that shaped the book into its final form, helping immeasurably in the difficult task of explaining for the non-scientist the complex scientific and technical arguments underlying the events described. Thus, though the responsibility for the accuracy of the facts and their interpretation must remain mine, whatever success this book may have in clarifying the

nature of the scientific problem and the dangers arising from the misuse of nuclear radiation will be to a large extent a reflection of his efforts.

Among the many others who contributed importantly to bringing this book into being, I must express my indebtedness to Larry Bogart, long-time conservationist and founder of The National League to Stop Environmental Pollution, who together with Leo Goodman of the United Automobile Workers was responsible for first drawing my attention to the full hazard of an unchecked nuclear technology.

In these researches, I was greatly aided by two of my colleagues, Dr. Donald Sashin and Ronald Rocchio, who worked out the computer programs that made the analysis of the vast amount of statistical data possible, as well as by Michael Szulman, Diane Gaye, Mitchel Margolis and Debbie Conant, without whose dedication in patiently collecting and analyzing the data the task would have been insuperable.

In the collection of the basic data, I was also generously helped by a number of young volunteer student assistants who spent long hours in the library. My gratitude goes to all of them, and in particular to David and Harold Colker, Randolph Strothman, and Gary Harris, whose important contributions were particularly appreciated.

Last but not least, I must express my deep appreciation to my secretary, Judy Czachowski, who patiently suffered through the agonies of

gathering the data, preparing papers and repeated retyping of the manuscript, and without whose cheerful help and dedication the task would have been vastly more difficult.

The duty to endure gives us the right to know.

Jean Rostand

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I

Thunderstorm in Troy

On Monday morning, April 27, 1953, the small group of students in Professor Herbert Clark's radiochemistry class at Rensselaer Polytechnic Institute walked into the metal shack that served as their laboratory, located high on a hill overlooking the city of Troy in upper New York State. The students set about making preparations for the day's experiments, but then Professor Clark interrupted to draw their attention to something unusual. All the Geiger counters were registering radiation at many times the natural rate.

Since instruments nearest the outer walls were giving the highest readings, several students immediately went outside with a portable Geiger counter. At once they found that wherever they walked, the count rate on the ground was far above normal, in some places a thousand times as high. In particular, beneath the spout of the gutters that carried the rainwater down off the roof of the shack, the needle gave a disconcertingly high reading. Evidently the previous night's heavy rains had brought down large amounts of radioactivity.

Dr. Clark quickly guessed the source. Such high readings could only have come from heavy deposits of fallout, the drifting clouds of radioactive debris created by the explosion of a nuclear bomb in the atmosphere. To verify his guess, he phoned John Harley, a friend and former colleague who now worked for the U.S. Atomic Energy Commission's Health and Safety Laboratory in New York City. As one of Dr. Clark's students recalled the story many years later, Harley's first reaction was that Clark must be kidding, and, expressing amused disbelief, he hung up. But a few minutes later, New York called back. Dr. Clark summarized the details of the morning's measurements: how the count rate from the gamma radiation on the ground was anywhere from ten to five hundred times normal, how the activity from beta rays had gone up even more, and how "hot spots" beneath rain-spouts and in puddles on the pavement showed still higher readings, much higher than he had ever observed after other nuclear tests, when it had been hard to measure any additional radioactivity at all. Thoroughly alarmed, the director of the New York Laboratory, Dr. Merrill Eisenbud promised to check personally into the situation, to send some of his top people to make their own measurements on the spot, and to take any steps that might be called for to protect the public health.

For, as Dr. Clark had just learned, there had indeed been a recent atomic bomb test, con-

ducted by the AEC in Nevada two days earlier. The bomb, code-named Simon and equivalent in power to 43,000 tons of TNT, had been detonated in the atmosphere some 300 feet above the desert. The upper portion of the mushroom cloud had reached an altitude of about 30,000 or 40,000 feet and then drifted 2300 miles across the United States in a northeasterly direction, passing high over Utah, Colorado, Kansas, Missouri, Illinois, Indiana, Ohio, and Pennsylvania before it encountered a severe thunderstorm in progress over most of upstate New York, southern Vermont, and parts of Massachusetts.

The storm was an extraordinarily violent one, accompanied by extremely high winds, hail, and torrential rains that flooded streets and basements, undermined foundations, and caused heavy damage to trees and houses. It was one of the heaviest flash storms Dr. Clark could remember. The sudden cloudburst, he surmised, had probably brought much of the fallout down in concentrated form. Dr. Clark quickly put his students to work in an effort to determine just how serious and widespread the danger might be.

Students set out with portable radiation detectors and began measuring the radioactivity on the pavement, on pieces of cloth, on asphalt roof shingles, on burdock leaves and other vegetation—any place it would be likely to collect and adhere. Samples were also taken of water from reservoirs and household taps. Within a matter of hours the students were reporting back

from nearby towns and cities such as Watervliet, Mechanicville, Saratoga Springs, Albany, and Schenectady that everywhere the radiation levels were about the same as on the campus. Typical readings were twenty to a hundred times normal, with hot spots up to ten times higher than that.

Now knowing the radiation levels as well as the source and age of the fallout, Dr. Clark could calculate that during the next ten weeks the total gamma radiation dose to the population from the radioactivity in the environment would be, on the average, roughly equivalent to that received from a typical diagnostic X-ray exposure. This was reassuring, since such a dose was not very different from what most people in the world receive each year from the naturally occurring cosmic rays that penetrate the earth's atmosphere. And it was well below the maximum permissible dose limits set by government agencies.

However, there was also the high radioactivity in the rainwater, which was certain to contaminate the reservoirs and thus the tap water. The samples of rainwater collected from a puddle on the campus had shown a radioactivity level of 270,000 micromicrocuries per liter, thousands of times higher than the maximum levels then permitted by AEC standards, which were set at 100 micromicrocuries per liter. Normal drinking water usually had an activity of about 1 micromicrocurie per liter.

There was, accordingly, much apprehension

among the students until the samples of actual drinking water from the taps and reservoirs could be analyzed early the next day. When this was done, the first of the tap water samples, taken Monday night, showed an activity of 2630 micromicrocuries per liter—not as great as was feared, yet still well in excess of the limit. But by that evening, the same tap gave a sample with a greatly decreased activity of 1210 per liter, while samples from nearby Tomhannock Reservoir ranged from 580 to 960. The radioactive rain was evidently becoming heavily diluted in the reservoir before reaching the taps in the households of Troy.

Thus, all concerned were greatly relieved that the total radiation doses received by the populace would probably turn out to be relatively small. It would not be necessary to filter the drinking water or decontaminate the streets and rooftops by means of elaborate and costly scrubbing procedures, a monumental task in view of the tenacity with which the radioactivity had been found to cling to rough surfaces such as pavement, asphalt shingles, and burdock leaves, and especially to porous materials like paper and cloth. Dr. Clark and his students found that even treatment with hot, concentrated hydrochloric acid—an extreme method—was only partially effective in removing the radioactivity from the objects to which it clung. The class also conducted tests to determine the strength of this radioactivity. Surprisingly, they found that it was comparable

to that reported the previous year by the AEC's New York Laboratory for fallout in desert areas only 200 to 500 miles from the point of detonation at the Nevada test site itself.

But the possible health effects of any internal doses that might result from eating, drinking, or breathing the radioactivity were considered negligible by the New York State Health Department and the AEC. And so it was decided that nothing further need be done. An editorial in the local newspaper expressed some concern, but soon the whole incident was forgotten.

Meanwhile, however, Dr. Clark, under contract to the AEC, continued to monitor the levels of radioactivity in the reservoirs, while AEC physicists, using an extremely sensitive gamma-ray detector mounted in an airplane, conducted extensive surveys of the entire region. Detailed reports on the findings were written by the staff of the New York Lab, but, since they were classified "secret," the public never learned of their contents. All that appeared was the following brief statement in the 14th Semi-Annual Report of the Atomic Energy Commission for the first half of 1953:

After one detonation, unusually heavy fallout was noted as far from Nevada as the Troy-Albany area in New York. Following a heavy rain in that area on the second day after the detonation, the concentration of radioactivity was from 100 to 200 curies per square mile. It

is estimated that this level of radioactivity would result in about 0.1 roentgen exposure for the first 13 weeks following the fallout. The exposure has no significance in relation to health.

One fact the AEC did not announce, and the general public did not learn, since it was later published by Dr. Clark in the obscure, highly specialized *Journal of the American Water Works Association*, was that, as the AEC continued its nuclear testing in Nevada during the spring of 1953, further rainouts repeatedly raised the radioactivity in the reservoirs serving Troy to levels comparable to those measured by Dr. Clark and his students the morning after the "Simon" rain-out in April.

II

The Unheeded Warning

The Troy incident was easily forgotten because, at the time, little was known about the effects of low-level radiation—either from fallout or from other sources. The subject had hardly even been thought about. Scientists generally assumed that such levels were harmless, since they produced no immediately observable effects. During the next few years, however, tremendously improved radiation measurement techniques coupled with detailed laboratory studies revealed many previously unsuspected hazards from fallout. And with these discoveries, the forgotten incident in upstate New York re-emerged and took on great significance.

By 1953, it was already known that many of the radioactive elements (called isotopes) created by an atomic explosion, once they entered the atmosphere in the form of tiny fallout particles, would contaminate food, water, and air and thus find their way into the human body. What was not widely known, however, was the extent to which these isotopes became concentrated in various body organs. Inside the body, they behaved just like their nonradioactive natural coun-

terparts. The isotope strontium, for instance, which is similar to calcium, settled in bones and teeth. Radioactive iodine behaved like regular iodine, seeking out and concentrating in the thyroid gland, an organ which is vital in regulating the growth and functioning of the human body.

It was in the case of iodine that some of the most alarming discoveries were made. In the early 1950s researchers found that iodine became concentrated in the milk of cows which grazed on pasture contaminated with fallout. When people drank the milk, the iodine began building up rapidly in their thyroid glands. Since the thyroid gland is small in size, the concentration was very heavy. Measurements revealed that in any given situation the radiation dose to the adult thyroid would be as much as a hundred times the external dose from the fallout in the outside environment. But far more important were the results of extensive studies conducted at the University of Michigan and published in 1960. These showed that the radiation dose to the thyroids of unborn children and infants was ten times higher than that to the adult because of the greater concentration in the smaller thyroids. This discovery held serious implications for the health of the children of Troy. It meant that the doses to their thyroids might have been as much as a hundred to a thousand times higher than those estimated by Dr. Clark and the AEC scientists, who had only considered the overall dose from the fallout in the external environment.

However, by the time these discoveries became widely known, a voluntary halt in atmospheric testing had been agreed upon by Russia, the United States, and England, and there was considerable hope that incidents of heavy fallout would never occur again. Thus it seemed less urgent to pursue investigations into the problem. But in 1961, during the Berlin crisis, Russia's detonation of a 100-megaton hydrogen bomb high over Siberia marked the resumption of large-scale atmospheric testing by the nuclear powers, and the levels of radioactivity in air and water once again rose sharply throughout the world. In the weeks that followed, an enormous peak of radioactive iodine was detected in milk throughout the northern hemisphere. As the testing continued, many scientists began to feel it was imperative to find a conclusive answer to the question: Just exactly how harmful was low-level radiation from fallout?

It was in this context that the well-known nuclear physicist Ralph Lapp wrote an article for *Science* magazine in 1962 which first focused attention on the significance of the Albany-Troy incident. Lapp's article showed that radiation doses far larger than those permitted by federal safety guidelines must have been received by the children of Troy and numerous other cities which had been subjected to similar "rainouts" in the early years of testing. The purpose of the article was to point out that the Troy incident provided an excellent opportunity to find out just what

the effects of fallout were. The surrounding area's population of half a million persons was large enough to insure that any increase in the normally low incidence of such radiation-caused diseases as thyroid cancer or childhood leukemia would show up. (The normal incidence of leukemia among children under ten years old was about two to three cases per year per 100,000 children. Thus, if any area with only a few thousand children were studied, no cases at all might be found in some years, even if the radiation were strong enough to double the normal expected number.) And the detailed radiation measurements taken by Dr. Clark's students and the AEC meant that relatively accurate estimates could be made of the doses involved.

The study which Lapp proposed had enormous potential ramifications. At the time, many people in government, military, and scientific circles still believed that mankind could survive the levels of fallout that would result from a nuclear war, levels thousands of times greater than those from peacetime testing. The United States had embarked on an extensive civil defense program based on this belief. But if it were shown that peacetime fallout levels led to a significant increase in fatal diseases, then by implication nuclear war would probably mean the end of mankind, and thus the vast nuclear war machinery developed by the United States and Russia would become useless. In the second place, if it were shown that large numbers of children had

already died from the effects of fallout, then tremendous public revulsion would probably be generated against *all* activities which released more radioactivity into the environment. These would include not just the testing of nuclear weapons, but also the monumental program planned by many governments and industries throughout the world for the peacetime uses of atomic energy. For nuclear power reactors, atomic gas-mining explosions, and other forms of nuclear engineering all normally release low levels of radioactivity and, in the event of an accident, they entail the risk of much worse. And, finally, those individuals who had been in positions of responsibility would have a terrible guilt to bear for the damage already done.

The appearance of Lapp's article also served to highlight another extraordinary fact. It was then seventeen years since the first atomic explosion at Hiroshima in 1945, yet no large-scale cancer studies such as he proposed had ever been carried out, even though the AEC had long been in possession of detailed fallout data for many areas of the U.S. A great deal of information existed on the effects of high doses of radiation, such as those received by the survivors of the explosions at Hiroshima and Nagasaki, but there was no real evidence regarding low-level effects, either from laboratory animal studies or from direct observations of large human populations. The lack of animal studies was somewhat understandable, since no such experiments could be

carried out at the extremely low doses produced by fallout without requiring hundreds of thousands or even millions of animals and many years to detect the small increases of a rare disease such as leukemia. But in the case of humans, such a large study population had already been created by the fallout from years of atomic testing. Yet the AEC had ignored this opportunity to resolve such an important issue. Thus, those who wished to minimize the danger of continued atomic testing could argue, in the absence of data to the contrary, that long-term, low-level exposure such as that from fallout had not been proven to increase fatal diseases.

The absence of such studies was all the more striking because there were already strong indications that such danger existed. It was toward the end of 1955 that Dr. Alice Stewart, head of the Department of Preventive Medicine at Oxford University, first became aware of a sharp rise in leukemia among young children in England. A young statistician in her department, David Hewitt, had discovered that the number of children dying of this incurable cancer of the blood had risen over 50 percent in only a few years. In the United States an increase about twice as large had occurred. One aspect of this rise was extremely puzzling: The leukemia seemed to strike mostly children over two to three years of age—there was little or no increase for younger children. This had not been the situation prior to World War II, when both groups had shown a

parallel, much more gradual rise. The question was: What new postwar development could be responsible for the increase in deaths among the older children?

Dr. Stewart undertook a study to find out. With the assistance of health officers throughout England and Wales, she obtained detailed interviews with the mothers of all of the 1694 children in those countries who had died of cancer in the years 1953 to 1955, as well as with an equal number of mothers of healthy children. By May 1957, the analysis of 1299 cases had been completed, half of which involved leukemia and the rest mainly brain and kidney tumors. The data showed that babies born of mothers who had a series of X rays of the pelvic region during pregnancy were nearly twice as likely to develop leukemia or another form of cancer, as those born of mothers who had not been X-rayed. As Dr. Stewart noted, the chance of finding such a two to one ratio purely as a result of statistical accident was in this case less than one in ten million. Thus, in the paper she published in June 1958, Dr. Stewart concluded that the dose from diagnostic X rays could produce a clearly detectable increase in childhood cancer when given during pregnancy.

This was an extremely low dose. It was roughly comparable to the dose that most people receive in only a few years from natural background radiation. (Mankind has always lived with a "natural background" of radiation, pro-

duced by cosmic rays and various naturally occurring radioactive substances. The annual dose from the radiation averages about 100 millirads.*) But still more significant, this dose was comparable with what the pregnant mothers of Albany-Troy must have received from the fallout of the "Simon" test in 1953.

In this connection, there was another finding of Dr. Stewart's study that was even more disturbing. This concerned the timing of the X rays. Children whose mothers were X-rayed during the first third of their pregnancy were found to be some ten times more likely to develop cancer than those whose mothers were X-rayed toward the end of pregnancy. In other words, the earlier the worse. This finding had much more serious implications for fallout than for medical X rays. Almost 90 percent of pelvic X-ray examinations occur shortly before delivery time, but, since fallout comes down indiscriminately on whole populations, it irradiates unborn children at all stages of development, including the earliest. The fallout hazard was further compounded by the tendency of various radioactive elements, such as iodine and strontium, to concentrate in vital body organs. This meant that the doses to the thyroids and bone marrows of unborn children

*The rad is the standard unit for measurement of radiation exposure. A millirad is one-thousandth of a rad. The federal government's maximum permissible yearly radiation dose for any single individual is one-half of a rad, or 500 millirads.

from fallout could be many times higher than the doses received from diagnostic X rays by the children in Dr. Stewart's study, which had already nearly doubled the cancer incidence.

But in order to establish a clear cause-and-effect relationship between the X rays and the additional cancer deaths, there had to be a direct relationship between the amount of radiation received by the fetus and the chance that the child would develop cancer a few years later. And indeed, when Dr. Stewart and David Hewitt examined the available records for the number of X-ray films taken, they found that there were distinctly fewer cancer cases among the children whose mothers had only one X ray than among those who had four or more. The number of cases where this information was available was too small to establish a conclusive connection between dose and cancer risk, but there was other evidence that supported this general trend. For example, whenever the X rays had only been taken of other parts of the body, such as the arms and legs, so that only a small quantity of scattered radiation reached the unborn child in the womb, the increase in cancer risk was only about one-fifth as great as in those cases where the abdominal region itself was X-rayed.

These latter observations were in direct contradiction to a belief that was essential to the continuation of all programs for nuclear testing and the peaceful uses of the atom—namely, the so-called “threshold” theory. This theory held that

there was a certain low level of radiation exposure, a "threshold," below which no damage would be caused. If this threshold was about the same as the yearly dose from background radiation or from exposure to typical diagnostic X rays, as various supporters of nuclear programs maintained it was, then there would theoretically be no ill effects from past or present weapons tests, from the radioactive releases of nuclear reactors, or even from the radiation persisting after a nuclear war, since this radiation would probably not exceed the threshold if it were averaged out over a lifetime. But Dr. Stewart's study implied that if there were any safe threshold for unborn children and infants it would have to be less than the dose from a single X-ray picture. And her finding that the risk of cancer seemed to be directly related to the size of the dose suggested that there might not be any safe threshold at all, and that *any* increase in radiation exposure might produce a corresponding increase in the risk. Even if the risk for a certain tiny amount of radiation was extremely small, say, one chance in ten thousand, then if millions of people were exposed to this radiation, hundreds would be likely to get cancer. Fallout had already exposed millions of people to doses comparable to those received by the children in Dr. Stewart's study, and the proliferation of nuclear explosions for peaceful purposes would make this exposure even more extensive.

There was widespread refusal to accept the

implications of Dr. Stewart's work. Her findings were regarded as doubtful for such reasons as their dependence on the memories of the mothers as to the number of X-ray exposures received. Other studies were cited which showed no effects from X rays. It was said that her study was inapplicable to fallout because it had been shown that a specified dose of radiation given all at once—as is the case with a diagnostic X ray—is more damaging than the same total dose given gradually over a period of weeks, months, or years—as is the case with fallout.

This argument opened up another important area of disagreement about radiation dangers. Were the cancer-causing effects of radiation cumulative? Or did body cells recover? There was no question that body cells did repair themselves in the case of such damage as radiation burns, which healed with the passage of time. Supporters of the threshold theory hypothesized that this would also hold true for cancer. This was another bulwark of the "threshold" theory, for, if such recovery did take place, then there would indeed exist a level of radiation low enough so that the body's repair mechanisms could keep pace with the damage.

However, evidence was soon forthcoming that would refute the criticisms of Dr. Stewart's study and thereby cast further doubt on the validity of the threshold theory. After the publication of Dr. Stewart's results, Dr. Brian MacMahon of the School of Public Health at Harvard University

undertook another study of the relationship between diagnostic X rays and childhood cancer. He constructed this study so that there would be no question as to the number of X rays given to the mothers. Using the carefully maintained hospital records of 700,000 mothers who delivered their babies in a series of large hospitals in the northeastern United States between 1947 and 1954, he compared the risk of cancer for the children of the 70,000 mothers who had received one or more X rays with the risk for the children of the remaining 630,000 mothers who had received no X rays during pregnancy.

The results of his study, published in 1962, fully confirmed the findings of Dr. Stewart: There was a clear and highly significant increase in the risk of cancer for the children who had been X-rayed before birth, and, most important, the risk did indeed increase with the number of X rays taken. The overall risk was somewhat smaller than had been found for the British children by Dr. Stewart, but this could easily be explained by the fact that the dose to the mothers in MacMahon's study from each X-ray picture was substantially lower than for those in Dr. Stewart's, due to improvements in X-ray technology. As for the studies cited by critics which did not show any increase in cancer risk from prenatal X rays, it developed that these were all based on small study populations, and even then the indications were that if these results were

carried out to larger numbers they would confirm Stewart and MacMahon.

But there was still one major question that remained unanswered. To what degree were the effects of diagnostic X rays comparable with those of fallout?

There were already many indications that the effects might be similar. Among these was the fact which had prompted Dr. Stewart to undertake her study in the first place, namely, the evidence that in both the United States and England cancer and leukemia among school-aged children had increased sharply beginning a few years after World War II. This was the period when nuclear fallout was first introduced into the atmosphere. And now, Dr. Stewart's and Dr. MacMahon's studies had served to point up the following significant aspects of this increase:

First, the effects of X rays, although very real, were not strong enough to have caused all of the very large general increase in childhood cancer, which ranged from 50 to 100 percent. Dr. Stewart herself estimated that X rays could only have accounted for perhaps 5 percent of this increase.

Second, this general increase had taken place only among children older than two or three—exactly the age group that had suffered the greatest effects from X rays. This suggested that some other form of radiation might be causing the unexplained portion of the increase, since the characteristic age at death was the same.

Third, other possible factors such as the introduction of new drugs, pesticides, or food additives had been ruled out because these factors had been found to be essentially the same for the healthy and afflicted children alike.

But the main reason why it seemed that fallout was at least as effective as X rays in producing childhood cancer was the growing evidence for a direct relationship between the number of X-ray pictures taken and the risk of cancer. For if the risk increased with each additional picture, as the studies of Stewart and MacMahon indicated it did, then this clearly implied that there was no significant healing of the damage and thus that the cancer-causing effects of radiation were cumulative. This would mean that the effects of a dose received over a period of time from fallout would be similar to those from an equal dose received all at once from X rays.

Such a direct connection between the amount of radiation absorbed and the likelihood of cancer could be predicted on the basis of a theory developed by Dr. E. B. Lewis of the California Institute of Technology. According to Dr. Lewis, cancer could be triggered if one particle of radiation scored a single bulletlike hit on a crucial DNA molecule in the chromosomes of a cell. The DNA contains the genetic code which controls the functioning and reproduction of the cell. If it were damaged by a particle of radiation, this might disrupt the governing mechanism and

cause the cell to begin the unlimited growth which characterizes cancer.

The significance of this theory was twofold. First, it was already established that such damage to the DNA was one of the ways that radiation produced hereditary or genetic damage in the female ova and male sperm cell—the type of damage that results in malformations and other harmful mutations in offspring. What Dr. Lewis stated, however, was that it was exactly the same type of damage, but to the DNA of any body cell, that could produce cancer. This was extremely important because it had already been decisively demonstrated that genetic damage was cumulative. In one experiment after another, using fruit flies and large colonies of mice, it was found that it did not matter how slowly or quickly a given dose of radiation was administered—in every case the number of defective offspring was essentially the same. The resulting effect on offspring was determined only by the total accumulated radiation dose received, regardless of the length of the time period over which it was given. There were some indications of repair in the ova of female mice, but the effect was relatively small at best. Thus there existed clear evidence that radiation effects of the type which produced genetic damage were cumulative, especially in the male sperm cell. But if Dr. Lewis was right, and radiation caused cancer in body cells in exactly the same way as it caused genetic damage in reproductive cells, then this

clearly implied that the cancer-causing effect of radiation was also cumulative. And if this was so, then the greater the radiation dose, the greater the risk of cancer. Dr. Lewis's theory therefore supported the findings of Stewart and MacMahon, and simultaneously gave weight to the theory that the cancer-causing effects of protracted radiation from fallout would be the same as for X rays given all at once.

All of this evidence combined pointed toward a single tragic conclusion: Man, especially during the stage of early embryonic life, was hundreds or thousands of times more sensitive to radiation than anyone had ever suspected.

III

A Small Error in the Assumptions

My own involvement in the subject of fallout hazards began in 1961. That was the year of the Berlin crisis, when Russia ended the voluntary moratorium on nuclear testing and the U.S. government called for a large-scale fallout shelter construction program. The intensified threat of nuclear war caused much concern in the scientific community, and in Pittsburgh, Pennsylvania, a group called the Federation of American Scientists, of which I was a member, decided to participate in a study of the chances for survival of a large industrial city like Pittsburgh in the event of a nuclear war. Since I was professionally involved in research on new techniques for reducing the radiation dose from medical X rays, and therefore was interested in the problem of low-level radiation effects, it was suggested that I join the section investigating the health hazards from fallout.

Almost as soon as we had begun our work, a disturbing fact emerged. All the calculations made by government agencies as to the radiation protection necessary after a full-scale nuclear war were based on the assumption that the adult

could tolerate the enormous dose of 200 rads spread over a few days and as much as 1000 rads over a year. Apparently it had been decided by the government's scientific advisory groups that it was not necessary to take into account the long-range after-effects of radiation, either on the survivors themselves or on their offspring. Yet, as I well knew from my own research, the reason why so much effort was being spent to reduce the dose from medical X rays was because the doses of only a few rads per year received by radiologists in the course of their work had been found to significantly decrease their life spans, while among their children there had been a definite increase in congenital defects. Furthermore, if Dr. Stewart was correct, only 1 to 2 rads would double the chances of a child developing cancer when the radiation was received in the last few months of pregnancy, and only one-tenth of this amount might have the same effect when received in the first few months. Exposed to the radiation levels that would be present in the aftermath of a nuclear war, then, a great many children born in the years following could be expected to die of leukemia, cancer, or congenital malformations before reaching maturity.

Additionally, these doses of hundreds of rads which the government agencies considered tolerable were only estimates of the external doses from the fallout in the environment. The internal doses from the fallout particles concentrated

inside the body, which would be hundreds or thousands of times higher still, had not even been taken into account, although the knowledge necessary for calculating these internal doses was widely available.

It thus appeared that the chances for survival after a nuclear war were being presented to the public in a far more optimistic light than scientific evidence justified. And then, with the publication in 1962 of Lapp's article in *Science*, revealing the extremely high internal doses received by the children of Albany-Troy from the 1953 rainout, it became evident that the same held true for the health effects of peacetime fallout. Since by 1962 the intensive nuclear testing was filling the rains all over the world with radioactivity approaching the amounts that had descended on Troy, the number of children that could be expected to die as a result was very large.

I made an estimate as follows: According to figures presented at congressional hearings, the fallout from each 100 megatons* of hydrogen bombs tested would give an overall dose of from 200 to 400 millirads to every man, woman, and child in Europe, North America, and Asia. This was approximately the total megatonnage of the bombs already exploded in the latest test series as of the end of 1962, and 200 to 400 millirads was roughly equivalent to the dose from a pelvic

*One megaton is the equivalent in explosive energy to a million tons of TNT.

X ray. Thus, if there were indeed no difference in the effects of diagnostic X rays and fallout, one could expect as much as a 20 percent increase in cancer rates for those children born within a year after the recent tests. Since, at the time, about one child in a thousand normally died of cancer before reaching adolescence, and since four million children were born in the United States each year, then every year some 4000 children normally developed cancer. Therefore, a 20 percent increase would mean close to 800 additional deaths in the United States alone. For the rest of the world, the figure would be perhaps ten times larger, all as a result of only the most recent atmospheric tests.

And these figures did not even take into account the probability of much larger doses from local rainouts, where the fallout was brought down in concentrated form. In the case of Troy, the type of calculations made by Lapp indicated that an overall dose of anywhere from a few hundred to a few thousand millirads must have been received by the unborn children in the area, equivalent to a whole series of pelvic X rays. Depending on whether they were in an early or late stage of development at the time, their chances of developing cancer would have been increased 100 percent or more.

Thus it was clearly of the greatest importance to see whether the number of leukemia deaths among the children of Troy had in fact begun to increase a few years after the fallout arrived.

(A characteristic delay in the onset of the disease, when radiation was the cause, had been found by both Dr. Stewart and Dr. MacMahon and was also observed among the survivors of Hiroshima and Nagasaki, who began developing this fatal form of cancer some three to five years after their exposure.) Furthermore, it seemed imperative that the worldwide scientific community be made aware of the implications of the data of Stewart and MacMahon, and of the urgent need for large-scale statistical studies of populations exposed to fallout. Accordingly, by late fall, 1962, I had completed an article on the subject and submitted it to *Science* magazine. This seemed the most appropriate place for publication, since it was the official journal of the American Association for the Advancement of Science (AAAS), the country's largest scientific professional association, and as such was read by a large, interdisciplinary audience of scientists throughout the world.

This was, however, an inauspicious time for the publication of an article with such negative implications for nuclear warfare and peacetime testing. The Cuban missile crisis, which brought the world to the brink of nuclear war, had just passed, greatly increasing pressure for further development and testing of nuclear weapons. Therefore, in anticipation of possible publication difficulties, I decided to submit copies of the manuscript to a few noted scientists in the hope of gaining added support.

One copy went to Dr. Russell Morgan, chairman of the Department of Radiology at Johns Hopkins University and head of the National Advisory Committee on Radiation of the U.S. Public Health Service. Dr. Morgan was one of the country's most knowledgeable experts in the areas of X-ray technology and low-dose radiation effects. In his reply, he stated that the article brought into focus important implications of the work of Stewart and MacMahon that had not been fully recognized, and recommended that it should be published with only a few minor changes. Dr. Morgan also gave me his permission to refer to his statement if the paper had to be resubmitted to *Science* after an initial rejection.

Another copy went to Dr. Barry Commoner, professor of botany at the University of St. Louis and one of the founders of the Committee on Nuclear Information, a group which pioneered in the public dissemination of information on the effects of nuclear testing. Dr. Commoner said in his reply: "I believe that it [the article] represents a very important contribution to the subject. I hope that it will be published in *Science* just as it stands. . . . Your conclusion regarding the need for large-scale surveys of the incidence of leukemia and other forms of cancer is of great urgency."

The article, however, was returned by *Science*, accompanied by copies of two reviews and a letter of rejection from the editor, Philip Abelson.

Abelson was a physical chemist who had an extensive background in the nuclear field. For many years he had worked closely with Glenn Seaborg, later chairman of the AEC and president of the AAAS, on the development of processes for the production of uranium, and he was now a member of both the General Advisory Committee of the AEC and the Project Plowshare advisory committee. (Project Plowshare was the name given to the AEC's program for the development of peaceful uses for nuclear explosives.) In his letter, Abelson stated that he had reviewed the article himself and found that "there is not enough solid material to justify publication." He further expressed the opinion that "there is really no evidence of the functional relationship between the number of X rays taken and cancer mortality." This meant that he did not consider significant the indications in the work of Stewart and MacMahon that the risk of cancer increased directly with the increase in X-ray dose, indications which were in sharp contradiction to the threshold theory.

Upon examining the enclosed comments of the other two reviewers, who were nameless as is the custom, I found that one was completely negative, stating that the article presented "no new observation" and ignored studies which showed no effects from diagnostic X rays. The other reviewer, however, recommended publication. Apparently, then, it had been Abelson's opinion which weighted the scales in favor of rejection.

A few days after the article was returned, I received an unexpected letter from Dr. James H. Lade, special assistant to the commissioner for radiological health of the New York State Health Department. Since 1951 Dr. Lade had also been director of the department's Bureau of Medical Defense, a part of the state's extensive Civil Defense Program, which had carried out an "exercise" at the time of the Albany-Troy incident. As medical director, Lade had been one of those who participated in the decision that no health protection measures were necessary after the incident and that no ill effects were to be expected. His letter read as follows:

Dear Mr. Sternglass:

I have had an opportunity to review your interesting paper on "Ionizing Radiation in the Pre-Natal Stage and the Development of Childhood Cancer," and noted your reference to Ralph Lapp's paper on the Troy-Albany fall-out in 1953. We in this department have done a little investigation of the circumstances which obtained in the Troy-Albany area at that time and the number of cancer cases and deaths reported in the age group who were under two years of age at that time. You may be interested in the results of these investigations, summarized in my attached letter to *Science*.

Yours very truly,
James H. Lade, M.D.
Director

Lade, apparently, either had been the negative reviewer or had been consulted by him. The *Science* letter to which he referred had been published in the November 9 issue in reply to Lapp's article. In that letter Lade attempted to minimize the possibility of any radiation effects in the Albany-Troy area from the concentration of radioactivity in the milk by arguing that the cattle in the area had not been turned out to pasture until about May 12, 1953, or some two to three weeks after the fallout had arrived on April 25. He argued that in view of the seven-day half-life* of iodine-131, the radiation would have decreased to only one-fourth its initial intensity by the time the cattle were turned out, so that Lapp's estimates of the dose to infant thyroids were at least four times too high. Lapp had, apparently, been unaware of this factor when he made his dose estimates, but Lade failed to mention that even if Lapp's doses were reduced by a factor of four, they would still be vastly greater than the permissible limits set by government agencies. Lade's letter also did not take into account the dose from the longer-lived isotopes such as strontium-90, strontium-89, and barium-140, with half-lives of twenty-eight years, fifty

*The half-life of a radioactive isotope is the time it takes for the radioactivity to diminish to half its original intensity. The half-life of iodine-131 is 7 days, so it is termed a short-lived isotope. Strontium-90, with a half-life of twenty-eight years, is a long-lived isotope. The radiation from a short-lived isotope is much stronger because all of it is given off in a shorter time.

days, and thirteen days, respectively, which would certainly still be present two to three weeks after the fallout had arrived. Nor did he mention the information published by Professor Clark in the *Journal of the American Water Works Association*, namely, that in May and June, many weeks after the first rainout, fallout from additional tests repeatedly produced new levels of radioactivity comparable to those measured for April. This meant, of course, that Lapp's estimates, far from being four times too high, were actually much too low, for Lapp had based his estimates only on the April 26 fallout. Lade, as medical director of the Civil Defense group, worked intimately with the scientists from the New York office of the AEC who sponsored the measurements of radioactivity in the reservoirs of the area. Thus he presumably would have been aware of this circumstance.

Lade further argued that because of the heaviness of the spring rains, the radioactivity had soon been largely washed off the vegetables and pasture. But Dr. Clark and his students had found that even extreme chemical treatment was only partially effective in removing the radioactivity from the leaves and other objects to which it clung. Thus, ironically, the only effect of the rains of May and June would have been to bring down even greater amounts of fallout than had come down in April.

As a final point, Lade stated that a review of New York Health Department records indicated

that no cases of thyroid cancer had since developed among the children who were under two years of age in 1953. His letter closed with the remark that "it seems most unlikely that an event which has resulted in no increase of thyroid carcinoma during the ensuing nine years will lead to such an effect in the future." Yet it was common knowledge among specialists in the field that radiation-caused thyroid cancer generally takes ten to twenty years to develop. And Lade said nothing about any increase in the incidence of leukemia, which by this time would certainly be detectable.

It could thus be determined by someone with a scientific background that Lade's letter contained absolutely no evidence to support his conclusion that the Troy fallout had been harmless. But how could the general public ever guess? This was the voice of the New York State Health Department.

Within a few weeks, I resubmitted my article to *Science* together with a letter referring to the statement given me by Dr. Russell Morgan. Within less than a month, it had been accepted for publication.

IV

A Ray of Hope

By the spring of 1963, when the article finally appeared in *Science*, the levels of radioactivity in milk were reaching unprecedented heights all over the United States. Extreme concern was being voiced both by scientists and the general public regardless of the repeated reassurances by local public health officials and the AEC that no danger existed. The intensified pressure began to penetrate to the highest levels of government, and in June 1963 President John F. Kennedy announced that this country, Russia, and Great Britain had agreed to negotiate a treaty to end all atmospheric testing. He further stated that he had ordered an end to all such testing by the United States. In July, the President delivered an address to the nation in which he urged the ratification of the treaty by the U.S. Senate. In this address, he referred to the threat of fallout as follows:

. . . the number of children and grandchildren with cancer in their bones, with leukemia in their blood, or with poison in their lungs might seem statistically small to some, in comparison

with natural health hazards, but this is not a natural health hazard—and it is not a statistical issue. The loss of even one human life, or the malformation of even one baby—who may be born long after we are gone—should be of concern to us all. Our children and grandchildren are not merely statistics toward which we can be indifferent.

It appeared that the issues involved in the question of fallout hazards were at last receiving widespread public recognition. In Congress, hearings on low-level radiation effects were being held by the Joint Committee on Atomic Energy, and for the first time citizens' groups and private scientists with no government affiliations were invited to testify on this subject. In August I received a letter from John T. Conway, executive director of the Joint Committee, requesting my presence at the hearings. Leafing through the transcript of the first half of the hearings held in June, I found a reprint of my *Science* article. It was followed by a lengthy critique prepared by the AEC's Division of Biology and Medicine. On examination, the AEC critique proved to be strikingly similar in wording and theme to the negative review for *Science*. It denied that any effects from low-level radiation had been proven to exist, stating that I had ignored the studies which showed no effects. The AEC critique, however, went further and cited figures showing an actual reduction in childhood cancer rates

after the heavy atmospheric testing in 1957. But the figures were only for children who were less than two years old when they died, a peculiar restriction, since Stewart and MacMahon had specifically demonstrated that cancer caused by the irradiation of unborn children only showed up *after* the age of two. No figures for older children were given by the AEC.

The Joint Committee hearings opened on August 20, the same day that Dr. Edward Teller testified before the Senate against ratification of the Test-Ban Treaty. Dr. Teller, the renowned nuclear physicist who had been instrumental in the development of the hydrogen bomb, expressed in his testimony the main arguments advanced by the treaty's opponents. He raised the possibility of future military breakthroughs by the Russians, stating his conviction that in the 1961–62 test series they had acquired knowledge about missile defense that this country didn't have. "This is the main argument against the test-ban treaty," he said. "It weakens our defense, and as long as we have reason to distrust Soviet intentions, the weakness of our defense will invite attack. . . . I came to recognize many years ago that real cooperation between Russia and us in the near future is impossible." As to the possibility of danger from fallout, Teller flatly stated: "From the present levels of worldwide fallout there is no danger. The real danger is that you will frighten mothers from giving milk to their babies. By that, probably much more damage

has been done than by anything else concerning this matter." Teller also stated that the treaty would endanger the program for the peaceful uses of nuclear explosions in the Project Plowshare program. He expressed his belief that such explosions could be carried out in "a very clean way," so that fifteen minutes afterward people could walk in the resulting crater "without exposing ourselves to more radiation than we have taken year in and year out in our laboratories."

At the same time as Dr. Teller was making this statement, in another part of the capital Dr. C. W. Mays of the University of Utah was testifying at the Joint Committee hearings about the thyroid doses received by children in Utah from one of the AEC's Project Plowshare explosions the year before. As measured by University of Utah scientists, the doses ranged from 10 to 60 rads—anywhere from ten to a hundred times the government's maximum permissible yearly limit for that body organ.

The second day of the Joint Committee hearings, during which I was present, began with the testimony of Dr. Shields Warren, who represented the National Academy of Science's Advisory Committee on the Biological Effects of Radiation. Dr. Warren had been the first director of the AEC's Division of Biology and Medicine, and as such he had been responsible for all of the AEC's early research and planning relating to the health effects of fallout. He was also head

of the U.S. delegation to the United Nations Scientific Committee on the Effects of Radiation. Dr. Warren began his testimony with a review of the effects of radiation on man, animals, and plants. The lowest dose listed on his chart was 0.001 rad, or 1 millirad, and this was followed by 0.01 rad, which he indicated as giving "no detectable effect." The next highest dose was 1.0 rad, again with the legend "no detectable effect." Dr. Warren evidently did not accept the evidence of Stewart and MacMahon which indicated that 1.0 rad received during pregnancy produced something like a 50 percent increase in the rate of childhood leukemia and cancer, clearly a "detectable" effect.

Nothing very serious appeared on Dr. Warren's chart until he got to 1000 rads. At 10 rads the legend was: "Barely detectable qualitative changes in lymphocytes," while at 100 rads the chart indicated only "mild acute radiation sickness; slight diminution in blood cell counts. Possible nausea and vomiting. . . ." Yet the studies of Hiroshima had revealed a doubling or tripling of leukemia deaths among the surviving adults exposed to this dose.

Only at 1000 rads did radiation actually appear lethal: "Depression of blood cells and platelets . . . death within twenty days." This was profoundly misleading: It was common scientific knowledge that 50 percent of all individuals exposed to 400 rads to the whole body

would die within a matter of weeks or months, while for 700 rads the figure was 95 percent.

For 10,000 rads, Dr. Warren's chart listed "Immediate disorientation and coma. Death within hours . . ." while "death of all living organisms" was reassuringly indicated as not occurring until 10,000,000 rads. In conclusion, Dr. Warren quoted the following passage from the United Nations 1962 report on fallout hazards, which he had helped to prepare:

It must be recognized that the human species has, in fact, always been exposed to small amounts of radiation from a variety of natural sources and that the present additional average exposure of mankind from all artificial sources is still smaller than that from natural sources.

This, then, had been the voice of the AEC's Division of Biology and Medicine, the National Academy of Sciences, and the U. S. delegation to the United Nations Scientific Commission.

When Congressman Melvin Price asked Dr. Warren what effects he would expect to observe in the 250,000 children in Utah who might have received 4.4 rads to their thyroids, as Dr. Mays had testified the day before, he replied: "I think it would be hard to find any effect, Mr. Chairman." But as Dr. Mays had suggested, if one does not look, one is not likely to find such effects, and indeed, no one had conducted the necessary studies of the children in Utah, in

Albany-Troy, or, for that matter, anywhere else in the world.

The next witness was Dr. E. B. Lewis of the California Institute of Technology, whose "single-hit," linear theory of cancer causation supplied an explanation for the findings of Stewart and MacMahon. This theory had formed the basis of my argument that the cancer-causing effects of radiation were cumulative and that doses received gradually from fallout would be similar in effect to those received all at once from diagnostic X rays. Dr. Lewis's latest evidence, based on a study of leukemia incidence among radiologists, indicated that low doses of radiation delivered over a period of many years could indeed lead to an increased incidence of leukemia, a finding which further buttressed his original theory.

There was now only one more witness scheduled to be heard before Dr. Brian MacMahon. This was Dr. Hyman Friedell, professor of radiology at Western Reserve University. The main burden of his testimony, as he stated, was to urge radiation protection agencies to set their standards on the basis of the linear theory of radiation damage, according to which there was no safe threshold of exposure.

Now Dr. MacMahon took the stand. He opened his testimony with an explanation of why certain other studies had not appeared to confirm Dr. Stewart's and his own results: "Every one of the negative studies has been based on small num-

bers, and in no instance do the results differ in a statistically significant degree from the expectation of a 40 percent increase in cancer risk among exposed children." Thus, in defense of his own work, Dr. MacMahon had effectively answered one of the major points in the AEC's critique of my article. He went on to summarize his results as follows:

I suggest therefore that the existing evidence is overwhelmingly indicative of an increase in cancer risk for children diagnostically exposed in utero; that this increase is seen for leukemia as well as for a variety of other individual forms of cancer; that the best estimate of this increase is that it is about 40 percent in excess of "normal" cancer mortality in the United States.

Turning next to the implications of his findings, he stated:

It seems to me that if this association is accepted, we must consider very seriously the possibility of cancer production by low doses of radiation such as encountered in X-ray diagnosis and even fallout.

As to the existence of a possible "threshold," Dr. MacMahon went on to say:

The exposure dose associated with pelvimetry [pelvic X rays] at the time of these studies is not known, but it was probably of the order of 2

to 3 rad [for a series of X rays]—substantially below the 50 rad that has frequently been mentioned as a possible threshold level. If a threshold for leukemia induction exists, then it must be below 2 rads. It would be a coincidence indeed if a threshold existed just below the dosage level at which studies have been undertaken.

One other major issue remained to be discussed, and that was the evidence for an increase in cancer risk with an increase in dose—termed the “dose-response” relationship. To this point, MacMahon now addressed himself as follows:

Furthermore, there is some indication in both Stewart’s and our own data that a dose-response relationship exists even within the low-dose range that is being considered.

Thus, MacMahon, together with all the other nongovernment scientists who testified in this part of the hearings, presented evidence in conflict with the AEC’s claims for the existence of a safe threshold.

Inevitably, Dr. MacMahon was asked his opinion of the use to which I put his data in my *Science* paper. He summarized his reply as follows: “. . . the argument used by Dr. Sternglass does not convince me personally, but, on the other hand, I cannot deny that ultimately his point of view may be correct.” He further stated, however, that he agreed with all the comments

in the AEC critique. This seemed puzzling, since I felt he had just answered the most serious points in his own testimony. It became clear that I would have to amass much stronger evidence before my hypothesis could gain the public support of many scientists.

After a brief recess, Dr. Eric Reiss was called to the witness stand. Dr. Reiss was a St. Louis physician and a co-founder, with Dr. Barry Commoner, of the St. Louis Committee for Nuclear Information. In his presentation on behalf of the St. Louis group, he concentrated on the subject of local fallout incidents resulting from the AEC's tests in Nevada. Referring to the many reports on local fallout prepared by the AEC, he pointed out that the general conclusion in all these official reports had been that "the test program has been carried out without any discernible threat to the safety of the local population."

"In contrast," Dr. Reiss went on to say, "our analysis of the same monitoring data published by the AEC shows that as a result of nuclear tests at the Nevada Test Site, in the period 1951-62, a number of local populations, especially in Nevada, Utah, and Idaho and probably other communities scattered throughout the continental United States, have been exposed to fallout so intense as to represent a medically unacceptable hazard to children who may drink fresh locally produced milk." He explained that the discrepancy between the AEC's conclusions and those of the St. Louis group arose out of the fact that

the AEC only measured the external dose from the fallout in the environment, while his group also measured the internal dose from the fallout particles concentrated inside the body, as Dr. Lapp had done in the case of Troy.

Dr. Reiss proceeded to present figures indicating that for 10 rads to the thyroid approximately one in 2860 children might be expected to develop thyroid cancer. Due to the long latent period for this disease, he explained, the damage would not show up for many years, possibly ten to twenty. He then cited estimated thyroid doses of 3 to 18 rads for the large population of Salt Lake City, Utah, as a result of fallout on May 7, 1952, and 5 to 40 rads for Albany-Troy on April 26, 1953 (this did not even include the much heavier fallout in June). Such hazardous doses, Dr. Reiss pointed out, arose not only from atmospheric tests but also from certain underground tests: "Venting [leaking of radioactivity] has been reported for at least seven underground nuclear tests," said Dr. Reiss, adding that his group was able to calculate, for example, that "fallout from the underground 'Gnome' shot delivered sufficient fallout to the vicinity of Carlsbad, New Mexico, to cause thyroid dose levels of from 7 to 55 rads to children."

And then Dr. Reiss turned to an astonishing aspect of the whole problem: If the dangerous internal doses had been estimated at the time, "simple preventive measures could have been taken to avoid exposure." It would only have

been necessary to warn the inhabitants of the area to avoid drinking fresh milk produced locally until the iodine levels died down. "We know of no instance," said Dr. Reiss, "in which such a warning was issued until the summer of 1962, when high iodine-131 levels observed in commercial milk supplied in Utah led state health officials to divert current milk from the market." And finally, Dr. Reiss noted that correct estimates of internal doses should have been possible by as early as 1954 *on the basis of the then-available scientific theory.*

The questioning of Dr. Reiss was nearing its end when John Conway asked him why his group had made only theoretical calculations of the thyroid doses without benefit of any direct measurements of the iodine levels in the milk. "I am delighted you brought that up," Reiss answered, "because it is the nub of the problem. If measurements were available, we would obviously have used them. The question is, why were no measurements made or reported publicly by those who had been charged with the protection of our health and safety?"

That evening the *Washington Post* gave prominent coverage to Reiss's testimony, as well as to the testimony given at the Senate's test-ban ratification hearings. At those hearings, Dr. John S. Foster, Jr., the new director of the AEC's Livermore Laboratory where the hydrogen bomb had been developed, had offered his opinion on fallout hazards:

You are asking for a quantitative answer which involves the understanding of an effect, namely the effect on the human being that is so small compared with his [radiation exposure from] background that we have not been able to measure it. One way to look at it would be to say that the fallout from all past tests affecting man for the next 50 to 100 years would be something like the same thing as deciding to live a few hundred feet higher up, higher above sea level, that is what it means.

Senator Robert Byrd of West Virginia then asked Dr. Foster what importance should be attached to the public's fears of fallout. Byrd phrased his question as follows: "If I may pose a hypothetical question, are you saying, in essence, that if you were a senator with the knowledge that you possess in the scientific field . . . you would disregard entirely this factor [public opinion] in your reaching a decision?"

To which Foster replied: "That is correct, sir, although it would be a very difficult position for me to put myself into. . . ."

The third day of the Joint Committee's low-level radiation hearings opened with my testimony. I reviewed all the evidence regarding low-level radiation effects on unborn children, presented my arguments as to the probable similarity in effects of X rays and fallout radiation, and gave estimates of the increase in childhood cancer which might be expected from nuclear testing. Then the questioning period began.

Congressman Craig Hosmer asked me how there could be any evidence in Stewart's and MacMahon's studies for a dose-response relationship when the radiation dose from hospital X-ray machines was known to differ widely from hospital to hospital, due to variations in the adjustment and quality of the machines. Such differences would mean that some mothers might have received the same or greater dose from a single X ray as others received from two or even three. Therefore, the finding of Stewart and MacMahon that there was an increase in cancer risk with the number of X rays taken might not indicate a dose-response relationship. I explained that it was also true, however, that in the case of any one machine, the mothers who received two X rays from it would definitely have received twice as much radiation as those who were X-rayed only once. Since the ratio of single pictures to double or treble pictures taken was generally the same for each X-ray machine, there could be no question that, on the average, the mothers who received two X rays had received twice as much radiation as those who received only one.

Congressman Hosmer next asked why one couldn't test the validity of the dose-response relationship simply by examining the statistics for leukemia in children after the test series in the early 1950s. I pointed out that this was precisely what Lapp, Mays, Reiss, myself, and others had urged the government to do without success.

The next question involved the principal argu-

ment used by the AEC to minimize the possible effects of fallout. Congressman Holifield asked whether I was aware that testimony had been presented by government scientists that fallout from all past testing had raised background radiation by only some 10 percent. Since the typical background dose was 100 millirads per year, this seemed to be only 10 millirads additional radiation, much less than the 200 millirads received from a modern pelvic X ray, or than the dose of 200 to 400 millirads that I had calculated for the recent test series.

What the spokesmen for the AEC and the Federal Radiation Council had done was to take the total radiation dose from all the bombs detonated so far—some 700 millirads—and average it out as if it were received uniformly over a seventy-year life span at a rate of 10 millirads per year. But in fact, most of the dose comes from the short-lived isotopes which predominate in fallout, and is thus delivered in the first few months after the tests. To illustrate this, I cited figures just published a few months before by the AEC's Brookhaven National Laboratory, located on Long Island, not far from New York City. According to these figures, during the first six months of 1963, the average radiation exposure on Long Island had risen to 4.7 millirads per week, as compared with a normal background rate of only 1.7 millirads per week. This meant that in only half a year the dose received by individuals in this area would be about 122

millirads as compared with a normal dose of only 44 millirads. Thus, instead of the negligibly small 10 percent increase which the AEC's seventy-year averaging procedure would give, the actual exposure to an infant in the womb would have been nearly triple the background rate. And this dose did not even include the effects of internal concentration. This, then, was the reasoning behind the reassuring statements of the world's radiation protection agencies that doses from atmospheric testing were negligible compared with all the sources of radiation in man's natural environment.

Congressman Hosmer next raised the point that the greater background radiation due to cosmic rays in an area of high altitude such as Denver as opposed to that in an area at sea level such as Long Island should lead to a higher leukemia rate in Denver. But in point of fact, as I stated, the difference in total background radiation between these two areas is actually quite small because most background radiation comes from sources other than cosmic rays. Cosmic rays contribute only about 30 millirads at sea level, compared to 50 millirads from the rocks and soil and 20 millirads from traces of natural radioactive substances in our body. Going to an altitude of 5000 feet increases the cosmic ray dose by only 40 percent, or a total of 12 millirads. Since place-to-place differences of 20 or 30 millirads in the radioactivity of rocks are common, the small difference due to altitude is

easily masked by this factor alone. Furthermore, cosmic rays do not concentrate in critical body organs as does fallout.

Thus, it would be nearly impossible to find differences in leukemia rates as a result of altitude differences even as large as those for Denver and Long Island, considering that MacMahon had to use a study group of some 70,000 exposed children to clearly establish an effect from about 1000 millirads of X rays. And I added that there are of course other factors beside radiation that enter into the likelihood of a child developing leukemia, making such a comparative study of the effects of slightly different background levels next to impossible.

Following up this point, Congressman Hosmer suggested that if factors other than radiation were involved in the cause of leukemia, did this not invalidate MacMahon's study? To this I replied that, in such a study, comparisons are carried out with control groups for whom all the other factors are, on the average, nearly the same as for those exposed to X rays, thus essentially eliminating the effect of such factors on the outcome.

After some discussion of the size of the dose from fallout, the chairman indicated that the committee's counsel, John Conway, had some points he wished to raise. After a lengthy prelude, Conway led up to the fact that the Federation of American Scientists' news release based on my paper said that 100 megatons of nuclear

fission energy had been released in the 1962 test series. Conway argued that since the amount released in 1962 was actually 76 million tons, then my estimate of the expected number of leukemia deaths was too high.

But it was clear from both my original paper in *Science* and my written testimony that the figure of 100 megatons applied to the entire test series that began in September of 1961 and ended late in 1962, not just to the amount detonated in 1962 alone. Thus, the news release, which was not even prepared by me, should have said more precisely, "1961-62 test series."

When this became clear, Chairman Price intervened to point out the simple nature of the misunderstanding, and when Conway nevertheless wanted to pursue the matter further Price thanked me for my testimony and called the next witness.

The debate over the test-ban treaty continued in Congress through most of September, and then the U.S. Senate voted overwhelmingly for ratification.

A highly revealing postscript to the entire test-ban treaty debate came to light in December, when the full proceedings of the Joint Committee hearings were published. Inserted in the record was a letter from Dr. Harold Knapp, a fallout specialist who had recently resigned from the AEC's Division of Biology and Medicine. In his letter, Dr. Knapp made reference to a 1957 incident in which a nuclear reactor in Windscale,

England, had gone out of control and emitted tremendous quantities of radioactivity into the atmosphere. The contamination from iodine-131 was so great that the crops and milk for hundreds of square miles around had to be seized and dumped. According to Dr. Knapp's letter, in 1960, when he was doing fallout research for the AEC, he came to the startling conclusion that in Utah "*depositions of iodine-131 per unit area at many inhabited ranches and communities must on several occasions have exceeded the maximum iodine-131 concentrations on pasturage found after the Windscale accident.*" (Emphasis added.)

As an example, Knapp cited an incident of relatively heavy fallout in the St. George, Utah, area on May 19, 1953. His calculations indicated that 24 hours after the explosion the iodine levels in milk must have reached 700,000 to 2,600,000 micromicrocuries per liter. He estimated that the thyroid dose for an infant who drank one liter (slightly more than a quart) of this milk each day for the three weeks following the test would be anywhere from 120 to 440 rads.

Knapp wrote a report containing these conclusions in 1960, but the AEC did not release his report for publication until August 16, 1963, just a few days before the second half of the Joint Committee hearings began, by which time it was evident that the independent scientists from Utah and St. Louis were going to make public their own similar findings. Thus, during the entire early effort to achieve a test-ban treaty,

this shocking and vital information was kept from the people and political leaders of the world, while hundreds of megatons of bombs were exploded by the U.S. and Russia during 1961-62. And since Knapp's report had not even been made public by the AEC and the Joint Committee by the time of the first half of the low-level radiation hearings in June of 1963, it had still been possible for the AEC to mislead Congress and the public until just before the end. From page 225 of the proceedings of the June hearings:

REPRESENTATIVE HOLIFIELD: And the testimony before this committee has been that tolerable permissible dose has not been reached by the amount of radiation that comes from man-made fallout radiation, is that not true?

DR. HARLEY: Yes, Sir.

Dr. Harley was John Harley, the former colleague whom Professor Clark had telephoned at the time of the Albany-Troy incident. At that time a scientist with the AEC's Health and Safety Laboratory in New York City, Harley had since become head of the lab, which had been responsible for the classified reports on the Troy fallout during the spring of 1953.

V

The Evidence Begins to Emerge

It was now an established fact that potentially dangerous radiation doses had been received by large numbers of people from fallout. What remained to be established was whether or not these doses had actually caused any damage. No direct statistical evidence had yet been gathered by either the AEC or federal and state health agencies, or, if it had been gathered, it had not been published.

This situation was abruptly transformed in March of 1964, when another letter by J. H. Lade of the New York State Health Department appeared in *Science*. This letter was written in response to repeated prodding by Ralph Lapp, who kept challenging the department to make public the actual figures after Lade had claimed in the September 1963 issue of *Science* that no increase in childhood leukemia had occurred as a result of the 1953 rainout. The exact words used by Lade in making this claim had been:

The cancer report files of this department reveal no increase in the incidence of cancer or leukemia over the past ten years in children of the

Albany, Troy, and Schenectady areas—who were 15 years or younger in 1963—as compared with children of this age elsewhere in upstate New York.

Lade's new letter contained the first detailed information on leukemia deaths among the children of the area. This was the data on which he had based the above conclusion.

The data consisted of a table which included all reported fatal leukemia cases among children under fifteen years of age in the Albany-Troy-Schenectady area between 1952 and 1962, together with the years in which these children were born. Examination of the table showed that, beginning in the fourth to fifth years after the 1953 rainout, the yearly number of reported leukemia cases quadrupled. This was strikingly similar to the delay in onset observed in the studies of Stewart and MacMahon and among the survivors of Hiroshima and Nagasaki. In the Troy area, after the eighth year, the number of cases began to decline once more, as they also did in the two Japanese cities. During the years 1952-55, before any radiation-caused leukemia cases would be expected to appear, there was a total of nine cases among children under age ten for whom the data was complete. During the years of expected peak incidence, 1959-62, there was a total of thirty cases. Statistical estimates demonstrated that the chances were less than

one in a billion that this increase of twenty-one cases could be regarded as purely accidental.

And when the cases were examined according to the year of birth, there was a very noticeable sudden increase in leukemia among the children born in 1953 and 1954, the period when the fallout radiation would have been strongest. When grouped in two-year periods, the figures in Lade's table for children under ten showed the following:

<i>Birth Years</i>	<i>No. of Cases</i>
1943-44	4
1945-46	5
1947-48	8
1949-50	5
1951-52	6
1953-54	13

How, then, had it been possible for Lade to claim that there had been "no increase in the incidence of cancer or leukemia over the past ten years in the children of the Albany, Troy, and Schenectady areas"? Upon rereading his exact statement, it became apparent that he had used a modifying phrase whose significance would easily escape the casual reader. This phrase was: "as compared with children of this age elsewhere in upstate New York."

But the "classified" measurements made by the AEC's New York lab had shown that the *entire upstate New York area* received heavy fallout

on April 26, 1953, and again during the next few months. Under these circumstances, there would of course be little or no *difference* in leukemia rates between Albany-Troy-Schenectady and elsewhere in upstate New York. Although he presented no data, Lade's statement clearly implied that in fact there had been similar increases in *all* these areas. (And indeed, later investigations revealed that there had been such rises all over New York State following three to five years after the onset of Nevada testing.)

In his 1964 letter, Lade presented a final argument to "rule out the fallout as a major factor in the leukemia incidence in that area for the 1953 births." According to his figures, there was also a large number of leukemia cases among the children born in 1956. Thus, Lade argued that the sharp rise in leukemia among those born in 1953 could not have resulted from the fallout, since there was also an increased incidence among the children born in 1956, when there presumably was no major fallout incident.

This piece of evidence seemed puzzling, especially since all the other data pointed so strongly to fallout. But then, a few months later, new light was thrown on the situation when the news of an extremely important study appeared in *The New York Times*. A group of investigators working with Dr. Saxon Graham at the Roswell Park Memorial Cancer Research Institute in Buffalo, New York, and at two other cancer research institutions, had found that the children of par-

ents who had been X-rayed as many as five to fifteen years *before* the mothers became pregnant also had a significantly increased likelihood of developing leukemia. This finding suggested that radiation could cause a type of *genetic* damage to the sperm cells of men and the ovaries of women which would make their children more prone to develop this disease.

In the case of Troy, then, it was indeed possible that such a genetic effect of radiation could lead to a greater incidence of leukemia among the children born in the years following.

There was another very important finding of the Roswell study: Children who had received multiple exposures from a series of diagnostic X rays to different parts of their bodies had a distinctly greater risk of leukemia than those who had only one part X-rayed. Fallout, due to the concentration of isotopes in various parts of the body, would lead to exactly this type of multiple exposure. Thus, in the Troy area, even children who were born *before* the rainout would be expected to show an increased risk of leukemia.

Independent studies had now demonstrated that radiation could cause leukemia in children prior to conception, in the womb, and after birth. What would the Troy data show if the cases were examined according to which of these three stages of development the children were in when the fallout arrived? When this was done, a fact of utmost significance emerged: Those children who had been in the womb or who were already

born at the time of the fallout were, on the average, one or two years older at death than the children who had been conceived afterward. This was exactly the same characteristic shift toward older age at death that had been observed by both Stewart and MacMahon in their studies of children X-rayed while in the womb, and it had also been observed in studies of leukemia among infants who had received medical X-ray treatment. It was as if the children of Troy had been marked with a sign. The fact that the children conceived after the fallout showed the normal earlier age at death was perfectly consistent, for it is generally acknowledged that there is a large genetic factor in all normally developing leukemia cases and that a significant portion of these cases are caused by genetic damage from natural background radiation. Thus the children whose leukemia had been caused by radiation damage to their parents' genes would be expected to show the normal age at death.

None of the data supplied by Lade was therefore inconsistent with the hypothesis that the fallout had caused the increase in leukemia in the Albany-Troy-Schenectady area. And this might well be true for other types of childhood cancer. Further investigation was definitely indicated. Since I anticipated difficulty in obtaining more complete and up-to-date data from Lade, I again contacted Dr. Russell Morgan, chairman of the U.S. Public Health Service's National Advisory Committee on Radiation. After reviewing

Lade's data and my conclusions, Dr. Morgan suggested that I use his name in writing to the New York State Commissioner of Health, Dr. Hollis Ingraham. In July I wrote Dr. Ingraham, asking for further information relating to cancer incidence in the Troy area and mentioning Dr. Morgan's concern about the data published by Lade. A few weeks later I received a letter from Lade indicating that he felt sufficient data was already available in his 1964 *Science* letter. It appeared that the New York State Health Department did not want to see the matter pursued any further.

VI

The Hidden Tragedy of Hiroshima

Lade's data on leukemia among the children of the Troy area, published in the expectation that it would disprove the existence of any effects from the 1953 fallout, now seemed to prove the very opposite. But there remained one major difficulty in accepting the evidence for an increase in leukemia, and that was the fact that the studies of the children conceived after the atomic explosions in Hiroshima and Nagasaki in 1945 had shown no such increase in leukemia, or in any other radiation effects, for that matter.

There had, of course, been very serious effects on the infants who were in the womb at the time of the explosions. A large fraction of these infants were lost through premature death, while among the few that survived more than a year, many suffered from congenital malformations and mental retardation. Of those unborn children who had been about one mile from the explosions and received estimated doses of 10 to 20 rads, nearly a third were found to have reduced brain size and mental retardation.

But among the many children *conceived* after

the explosions, there appeared to be no effects, as opposed to the situation in Troy.

Late in 1966, I was given a copy of the preliminary results of a Yale University study of Hiroshima and Nagasaki carried out by Dr. S. Finch and a group of colleagues and sponsored by the Atomic Bomb Casualty Commission. The study involved an examination of the incidence of leukemia among some 17,000 children whose parents had been within 2000 meters of the explosions. Radiation doses ranged from about 3 to 30 rads, with a few as large as a hundred. These were certainly much larger than the doses from typical pelvic X rays received by the mothers in the study by Dr. Saxon Graham, doses which increased the risk of childhood leukemia among the children conceived many years later by as much as 100 percent.

The control group used for comparison purposes in the Yale study consisted of the population in the suburbs farther than 2500 or 3500 meters from the explosion, where the radiation from the bomb was calculated to have been less than that from natural background. With such a tremendous difference in doses, there could be no question that there should be a substantial difference in effects. Furthermore, since there had been virtually no fallout in the two cities, the doses had been calculated from the instantaneous flash of the bombs alone. Such a flash was similar to the way in which the dose from diagnostic X rays is delivered, so, if anything, the

results of Dr. Graham's study should have been even more applicable to this situation than to Troy, where the fallout dose was delivered over a period of months or years.

Still another study of Hiroshima and Nagasaki had also found a similar lack of effect. This study had been conducted over a period of nearly two decades by Drs. J. V. Neel and W. J. Schull of the Department of Human Genetics at the University of Michigan. The puzzling nature of the results was emphasized by Dr. Neel himself during the course of a lecture series in 1963: "In view of the vast body of data regarding the mutagenic effects of radiation, it can scarcely be doubted that the survivors of Hiroshima and Nagasaki sustained genetic damage." To this he added: "The question is not 'Is there damage?' but rather 'Can the damage be detected?' "

It was clear that, unless this problem could be resolved, any evidence on the Troy incident would always be subject to this serious criticism. Accordingly, one day I decided to look up the original data on the radiation dose measurements for the two Japanese cities, collected by E. T. Arakawa of the AEC's Oak Ridge National Laboratories. And as I examined Arakawa's figures, I noticed that while there had indeed been little fallout in Hiroshima and Nagasaki proper, the fallout had drifted down on the suburbs a few miles away.

But if there had been fallout in the suburbs and beyond, and this fallout was far more damag-

ing to the ova, the embryo, and the infant than anyone had realized when the studies were set up, then the so-called nonexposed control populations beyond about 3000 meters, the groups used for comparison purposes, had in fact *also* been affected by the radiation. And just as in the case of Lade's comparison of the leukemia rate in the Albany-Troy-Schenectady area with that in nearby areas of upstate New York, a search for a difference would fail to indicate any effect.

The fallout doses were by no means small. Arakawa reported that in the western suburbs of Hiroshima, namely Furue, Takasu, and Koi, about 5000 meters from the blast, the external radiation dose from the fallout in the environment alone probably amounted to several rads—at least twenty times what Dr. Clark and his students were to estimate for the Troy population eight years later. In the case of Nagasaki, the fallout had been even heavier, giving external doses as high as 100 rads in the suburb of Nishiyama. This was a thousand times greater than in Troy. Such doses would greatly increase the leukemia rate for the children born all over the nearby regions that had been supposedly free of radiation, completely masking the effect if one looked only for a comparative difference, and not for an increase over the rate for preceding years. It seemed incredible that the possible effects of such large doses could have been overlooked

in the two major studies of the effects of atomic warfare, yet they had been.

A few months later, in the spring of 1967, after I had taken up my new position as professor of radiation physics at the University of Pittsburgh School of Medicine, I came across the evidence which confirmed this conclusion. It was contained in a paper published in the *New England Journal of Medicine*, bringing up to date all the findings on leukemia among the survivors of the two Japanese cities. One glance at the first figure told the story. As compared to the rate for the preceding years, not only had there been a rise in leukemia incidence among those who were less than 1500 meters from the explosions, but there was a similar though somewhat smaller rise for the population beyond 10,000 meters who could not possibly have received any of the direct radiation from the flash of the bomb. Both rises followed the characteristic pattern of radiation-caused leukemia. Furthermore, there was another sharp rise in leukemia for both of these groups in 1958, four to six years after the first large series of hydrogen bomb tests in the Pacific and Siberia in 1952-54, tests which resulted in heavy fallout in Japan and in contamination of the fish used widely as a staple item in the Japanese diet. Again the delay in onset coincided exactly with the established pattern of radiation-caused leukemia.

In fact, for all of Japan, leukemia rates rose sharply by 50 percent between 1946 and the

early 1950s, just as Dr. Stewart's statistician, David Hewitt, had originally observed for England and the United States. This was followed by another sharp rise as of 1959. And just as the rates had turned down again in Hiroshima and Troy, they declined again throughout Japan to half their peak intensities four to six years after the temporary moratorium of 1958-61, proving that these rises had not been due to such factors as improved diagnostic methods or increased use of medical X rays, as some had suggested.

The last major argument against the connection between the 1953 rainouts and the rise in leukemia in Albany-Troy had disappeared. By now, some three years had elapsed since my last attempt to get additional data on the Albany-Troy-Schenectady area from the New York Health Department. I sent another letter to Dr. Lade in a final effort to obtain more detailed and up-to-date information. Within a few weeks, the following reply arrived:

Doctor Sternglass, Sir:

I would be most willing to provide you with the data you request in respect to the occurrence of leukemia in children in the Albany-Troy-Schenectady area if there were any reason to suppose that they had sustained a significant exposure to fallout radiation. In my letter to the editor of *Science*, 141, 1109 (13 September 1963), I pointed out that children on a milk diet in this area at the time of 1953 could not

have had a significant exposure. I fail to see, therefore, how further data could be "valuable for our understanding of low dose rate effects."

JHL

Reluctantly, I began the preparation of a final paper for *Science*, based on the incomplete data from Lade's brief letter of 1964.

It appeared unlikely that *Science* would publish the Albany-Troy paper in the near future, so I also decided to present my findings at the forthcoming annual meeting of the Health Physics Society, to be held in Denver, Colorado, in June of 1968. This professional society had been founded in 1955 by Dr. Karl Z. Morgan and a few other physicists who, since the early years of atomic energy, had been concerned with the health aspects of this technology. The society was officially dedicated to "the protection of man and his environment from unwarranted radiation exposure." Over the years it had acquired many members who were professionally engaged in safety planning for nuclear weapons tests and nuclear industry activities.

Dr. Morgan, one of the most widely respected individuals in the health physics field, had himself become a controversial figure in recent years, due to his outspokenness regarding the widespread use of inadequate medical X-ray equipment and procedures. Long convinced that there existed no evidence for a completely safe "threshold" of radiation exposure, Dr. Morgan had

incurred heavy criticism from radiologists' associations and others for his estimates that thousands of additional deaths of unborn children were being caused each year by unnecessary over-exposure to medical X rays. Prior to the June meeting, I sent Dr. Morgan a copy of the paper I had just submitted to *Science* and received a letter from him expressing deep concern about the seriousness of the evidence, as well as indignation about the refusal of the New York State Health Department to provide the additional data.

At the annual meeting in Denver, the press was well represented, word having gotten around about the nature of my findings. I presented the evidence and concluded by urging that much more detailed studies be made of other areas known to have received heavy fallout. During the discussion period that followed, many of the questions dealt with the inadequacies of Lade's data—for example, the lack of such facts as the birthplace of each case and the particular type of leukemia involved. I could only point out that this information was not available to me.

Reports of the Troy findings were carried widely by the press both in the U.S. and around the world. Among the many phone calls which came during the aftermath was one from an Associated Press reporter in Detroit, who informed me that the New York State Health Department had just issued a news release stating that "there is no evidence to support a Pittsburgh

professor's report that radioactive fallout over the Albany-Troy, N. Y., area in 1953 has increased the incidence of childhood leukemia." The subsequent AP dispatch based on this release added that "Sternglass had consulted the department earlier and had been told the department's opinion." This "consultation," of course, had consisted only of requests for information that were repeatedly denied, while "the department's opinion" consisted of the arguments advanced by Lade in his correspondence to minimize the significance of the data. But it appeared likely that newspaper readers would gain quite a different impression.

It later developed that segments of the Health Physics Society were also displeased by the publicity. In a letter to the society's board members a month after the annual meeting, R. E. Alexander, chairman of the society's public relations committee, stated that some board members had complained that "the publicity about the paper of E. J. Sternglass . . . was damaging to the nuclear industry." After defining the "basic publicity objective" of the society, namely, "to let the public know that due to a frankly acknowledged need, we have a new technology, health physics, which will permit them to enjoy the benefits of nuclear energy safely," Alexander went on to say that "while we try to avoid publicizing papers that do not contribute to our basic objective, *there is no way to prevent such publicity absolutely.*" (Emphasis added.)

VII

Death Before Birth

A few months after the Denver meeting, a letter arrived from *Science* flatly rejecting the Albany-Troy paper. There were two reviews enclosed, both negative. One contained references to nonpublic data available only to the New York Health Department, indicating that the reviewer was a member of that agency. Apparently unaware of the irony, this reviewer claimed that "the number of cases in the Albany-Troy area used in this study are too few to warrant the conclusion." This was, of course, the same number of cases that Lade had used in his 1964 letter to *Science* to prove there had been *no* effect from the fallout, and his conclusion had then been deemed suitable for publication by the editor. And this very same data had also been cited frequently to the same end by the AEC and the New York Health Department. Apparently these figures were only adequate to prove there was no effect.

I now became more determined than ever to pursue the effort to have the full evidence on the worldwide effects of nuclear fallout on childhood leukemia exposed to the scientific com-

munity and the public at large. The issue was far greater than the rejection of just another scientific paper. The nature of modern science depends upon the free communication of even the most disturbing ideas, since only through their widest possible examination can the essential process of the gradual correction of errors be accomplished.

And so I began the task of answering each objection raised by the reviewers. One of the points was that the increase in childhood leukemia could probably be explained simply by the increase in the number of children born in Albany-Troy-Schenectady during the postwar years. It was really quite evident that this factor could not possibly begin to explain the fourfold rise in the annual number of cases in an area where the entire population had only increased by 10.5 percent from 1950 to 1960. But now this had to be proved in detail.

I began the laborious task of going through the volumes of the U.S. Vital Statistics, extracting the figures for annual births in the three counties of upstate New York for which Lade had given the leukemia rates. While copying down the numbers, I noticed that there was also a listing for fetal deaths—stillbirths, miscarriages, and other forms of death before birth. Remembering that an increase in fetal deaths had been looked for by J. V. Neel in his study of genetic effects on the survivors of Hiroshima and Nagasaki, I decided to note down these figures as well.

As expected, the gradual increase in annual births during the period from World War II to 1953 was far too small to account for the subsequent increase in the number of leukemia cases. The births had increased by only some 50 percent, while the leukemia cases had risen by over 300 percent.

But the figures for fetal deaths showed something quite unexpected. After steadily declining from 23.8 cases per thousand births in 1941 to a low of 14.4 in 1952, the number of fetal deaths suddenly refused to decline any further. It had only declined to 14.2 seven years later in 1960. This type of change is termed a "leveling off" in the rate of decline. In the case of fetal mortality, such a leveling off was most significant, for the original pattern of steady decline was the result of steady improvements in prenatal medical care which should have enabled the decline to continue until reaching an "irreducible minimum." This minimum had clearly not yet been reached, as was shown by the fact that the decline started again after each discontinuation of testing. More significantly, it reached an alltime low of 11.7 in 1965, two years after large-scale atmospheric testing had ended.

Could it be that this effect on fetal mortality had also been missed by Dr. Neel because his study had not taken into account the effects of fallout on the control population? Was it some kind of statistical illusion, or was it real, and

therefore far more serious than even the rise in leukemia?

The library closed for the day, cutting short my investigations. But the implications were staggering. For it was widely known among statisticians that the number of fetal deaths listed in the U.S. Vital Statistics was perhaps no more than one-tenth of the total which actually occurred, since many cases, especially in the early months of pregnancy, were never reported to the authorities. Before the fallout, there had been only two or three leukemia cases per year in the Troy area, as compared with some 150 to 200 reported fetal deaths. The actual number of fetal deaths in the area would probably have ranged from 1500 to 2000. If the relative increase in fetal deaths in Albany-Troy-Schenectady after the fallout was comparable to that for leukemia, then it would involve the loss of hundreds of times as many unborn children as might die from leukemia.

If it was true that fallout caused an increase in fetal mortality, then a similar effect should have occurred not only in the Troy area but also in the rest of New York State, parts of which had been exposed to various other rainouts throughout the 1950s. The next day I returned to the library and went through the data for New York State as a whole. Exactly the same pattern existed: There was a steady decline in fetal deaths toward a minimum of 22.2 per thousand births in 1950, but then the rate actually rose again to 23.3 in 1953 instead of continuing its decline to

the 20 per 1000 live births which was to be expected if the normal downward trend had continued. After a brief drop in the mid-fifties, fetal mortality rose once again following the major test series in 1957 and 1958, so that the rate exceeded the expected number by almost 50 percent in 1960. The gap widened steadily as all further improvements in living standards, diet, and maternal health care suddenly failed to have any further effect.

But when I reached 1964, I found the most extraordinary figure of all. In a single year the number of reported fetal deaths in New York State had jumped by 1500 cases. After this it declined once more—the exact same pattern as in the Troy area. In the single year 1964, then, there must have been some ten times this number of reported cases, or, ten to fifteen thousand additional children lost by miscarriage or stillbirth in New York State alone. This tremendous step-like increase for the entire state was clearly connected with the 1961–62 test series, from which large peaks in iodine and other short-lived radioactivity resulted when the spring rains came down in 1963. Unlike the local rainouts of the 1950s, the fallout from these extremely large tests came down much more uniformly over large areas of New York State. Thus there would have been few, if any, unexposed sections, and the state as a whole would show the kind of sharp increases that earlier had been seen only in localities like Albany-Troy.

But could this extraordinary figure be the result of some statistical fluctuation, or a sudden improvement in reporting methods which happened to coincide exactly with the period of the highest fallout levels ever recorded? There was one way to check this very quickly. Unlike the number of fetal deaths, more than 95 percent of all live births are reported to the public health authorities since nearly all of them take place in hospitals. So if there had really been an increase of some 15,000 fetal deaths in New York State in 1964, then there would have to have been a corresponding sudden drop in the number of children born live the following year. And this is exactly what happened.

For 1962, the total live births in New York State were listed as 354,152. For 1963, the number had increased to 355,760. For 1964, there was a drop to 351,602. But for 1965, there was a sudden decline to 335,628. This was a drop of 15,974 live births, or almost exactly the number of babies lost in 1964 through stillbirth or miscarriage. The rise in fetal deaths must therefore have been real.

It was imperative to make still another test. New York State in the early 1960s was more or less typical of the United States as a whole with respect to the levels of fallout in milk, food, and water. Therefore, the entire country should have shown the same effect: some ten to fifteen times as many fetal deaths in 1964, and a corresponding sharp drop in live births in 1965.

It took only a few minutes to find the figures for the children born live in the United States during these years. For 1964 the number was 4,027,000, and for 1965 it had declined to 3,760,000, a sudden drop of 267,000, the sharpest single decline in the entire history of the United States. And for the entire country the year before, fetal deaths showed a corresponding jump.

It seemed that if there had been about twenty times as many bombs detonated during the 1961-62 test series, there would probably not have been any children born live in 1965.

VIII

The Crucial Test

When I returned home from the library late that night, I told my wife what I had just learned and we discussed what should be done. Should I stop my efforts to publish the article on leukemia in Albany-Troy and concentrate all my energies on this apparently much greater effect of fallout? How could still more convincing evidence best be found? How could scientists and government officials be quickly informed of this discovery so that independent studies to check the findings could be carried out by others as soon as possible?

At that very moment, the French were continuing their atomic tests in the Pacific, while the Chinese were starting to detonate larger-than-ever bombs whose fallout was already drifting over the U.S. If the rise in fetal deaths had truly been caused by fallout, then a simple calculation showed that for each additional megaton of nuclear energy released some 2,000 to 4,000 infants would be stillborn in the U.S. within a year, and perhaps ten times as many all over the world. And this figure did not even include the many infants who would be born with congenital

defects or who would die of cancer and leukemia in the first years of life.

Yet what if my interpretation of the data was wrong? Should I spend many more months or even years gathering more and more detailed data on many additional populations, examining every conceivable alternative explanation for these effects, and only then publish the findings? Clearly this was not just another scientific study, to be handled with more deliberation than urgency. Here, every additional megaton bomb that the French or Chinese tested might mean the loss of thousands of babies, while every additional nuclear weapons system built would increase the certainty that human life would end if these weapons were ever used. The AEC was in the midst of plans to set off another large cratering explosion in Nevada within the next three months in an effort to prove the feasibility of excavating a new Panama Canal by means of hydrogen bombs. Such a test was bound to release large amounts of radioactive debris that would drift all over the United States and northern Europe.

On the other hand, many scientists would surely consider it irresponsible and alarmist to voice concern to the public before all the evidence had been gathered and submitted to the scientific community for detailed consideration. And there would of course be strenuous opposition from all the proponents of nuclear energy for military and peacetime use. Yet by now it was amply

clear that I could not expect quick publication of these findings in the widely read scientific journals, such as *Science*. At best, it might take a year to gather the necessary support and definitively overcome the objections of reviewers.

Ultimately I resolved to take the findings to the public if discussions with colleagues failed to reveal any alternative interpretations of the evidence. Meanwhile, I continued to work at the library, trying to confirm these incredible findings.

One test would be that in a state comparable to New York in economic level and quality of medical care, but where there had been less-intense fallout, the change in the infant mortality decline should have been correspondingly smaller. California, upwind from the Nevada test site, met these criteria, and so I plotted the data for that state. The fetal mortality rate did in fact continue to decline at an undiminished rate during the early 1950s—while New York State had already begun to show sharp rises—reaching rates almost half those in New York. Only in 1955 to 1958, after the large hydrogen bomb tests in the Pacific, did California also begin to show a noticeable leveling off in the rate of decline. But at no time did this leveling show the sharp rises observed in New York State that would be expected from the much more intense short-lived radioactivity produced by the relatively small “tactical” weapons tests carried out in Nevada.

But the most crucial test involved the deaths of infants. For some of my colleagues in the field of public health had pointed out that fetal mortality figures, being incompletely reported, are not nearly as reliable as those for infants who are born live but die before the age of one. And when I examined the infant mortality figures for New York City and New York State, I found that they did in fact show the same peaking in 1964 and 1965 as did the fetal deaths. Especially sharp was the increase in deaths among the infants under 28 days old, which are known to reflect most strongly any effects that occurred during embryonic life.

In order to tie these upward changes in fetal and infant mortality to fallout, it would be necessary to make a detailed comparison between the actual measured levels of radioactivity in the food and milk with the mortality rates to see if the rises and falls coincided. Fortunately, data on fallout levels had been made publicly available beginning in 1957 after a struggle between the Public Health Service and the AEC, which wanted to keep them classified. The first area to check would be New York, for which the fetal and infant death rates were already collected. At the outset, I did not know for certain which of the isotopes in the fallout were causing the principal damage. But then I saw that each time the levels of the short-lived isotopes, such as iodine-131 and strontium-89, shot up to their highest peaks, there was a sharp rise in fetal

mortality within a year. The first of these sharp general rises occurred after the very large Nevada, Pacific, and Siberian tests in 1957-58, and the second and highest took place following the tests in 1961-62, the high levels of short-lived isotopes in milk peaking in the spring and early summer of 1963.

Thus it appeared that, in the case of infant and fetal mortality, the short-lived isotopes produced especially by the smaller fission bombs were dominant. These isotopes gave off their radioactivity anywhere from ten to a hundred times faster than the long-lived strontium-90 and cesium-137, or than the carbon-14 and tritium produced by the hydrogen bombs. Inside the body, then, they would give the fetus the highest dose in the shortest time. Certainly iodine-131 could be a major source of damage. In a paper just published in January 1968, Merrill Eisenbud, who had been head of the AEC's New York Health and Safety Laboratory at the time of the Troy incident, reported an actual measurement of the iodine in the thyroid of a twelve-week-old fetus aborted in New York City in 1962, the peak year of testing. The fetus had received a thyroid dose ten times as large as that being received by the average newborn infant during the same period. Eisenbud gave this peak average thyroid dose for newborns in the city in 1962 as 200 millirads. So the thyroid of the fetus must have been receiving around 2000 millirads, or 2 rads, a truly enormous dose for this crucial

organ compared to the 75 millirads it would normally have received from natural background radiation. And this was just the dose to a single organ from a single isotope. The total effect on the developing fetus from the combined concentration of different isotopes in different organs must have been vastly greater. In this connection, Eisenbud also gave monthly figures on strontium-90 in milk for New York City, and these too showed sharp peaks in 1958, 1962, 1963, and 1964, corresponding to the years of greatest increases in fetal deaths.

But what would be the effects of the long-lived strontium on fetal and infant mortality? The intense radioactivity of the short-lived isotopes generally died out within six months to a year, but the strontium, with its half-life of twenty-eight years, would persist in the environment and in the diet, continuing to build up in the bones and genetic material of the exposed people until reaching an equilibrium level where natural metabolic processes would remove it at the same rate as it was being taken in. Studies had shown that this peak level was generally reached some four to five years after continuous intake began. Therefore, it seemed likely that any effects of strontium on infant mortality would parallel this accumulation in the bodies of the parents. If this was so, then after the initial sharp rise caused by the short-lived isotopes there would be a dip, followed by a gradual rise culminating in a second, broader, lower peak extending generally between the third

to fifth years. Thereafter, if no additional strontium was added to the diet, there would be a slow decline which would probably accelerate rapidly after a few years as the strontium in the environment was dissipated and diluted by natural processes.

When I discussed the findings with Dr. Barry Commoner, now at Washington University in St. Louis, he suggested that I compare the leveling of the decline in fetal and infant mortality with the measured amounts of strontium-90 on the ground and in the milk for different areas of the United States. It took only a few days to discover that the pattern followed closely the levels of strontium-90 that accumulated in the environment after the onset of hydrogen bomb testing in the early 1950s. Furthermore, the graphs consistently showed two peaks in tandem—a sharp peak within a year after each test series, when the levels of short-lived isotopes as well as strontium shot up, followed by a second slower rise culminating between three and five years later. The second peaks were especially high, probably because each of the enormous fusion bombs had actually produced hundreds of times as much strontium-90 as one of the earlier atomic fission bombs, even though the hydrogen bombs had been advertised as being much “cleaner.” For, as Ralph Lapp and the British physicist Joseph Rotblat had each discovered independently in 1954, in order to get a “bigger bang for a buck” as U.S. Secretary of

Defense Charles Wilson put it, Edward Teller and his weapons engineers had surrounded the hydrogen bombs with cheap, abundant uranium-238. As a result, the total explosive force could be doubled at no additional cost, but the levels of strontium-90 in the bones of living creatures were vastly increased.

The final task that remained was to make certain that there existed no other known explanation for the halt in the decline of infant mortality in the United States. Various of my associates in the University of Pittsburgh School of Public Health had said that to the best of their knowledge no other cause had been found. It was suggested that I review the results of an international conference on the problem, held in Washington, D.C., in May of 1965.

The summary of the conference revealed that extensive studies had been made comparing the U.S. with five European countries where there had been much less of a slowing in the infant mortality decline, or none at all. These studies had failed to find any explanation for the sudden worsening of the situation in this country. However, fallout had not even been considered as a possible factor.

According to the report, the U.S. infant mortality problem had become so serious since it was first noticed in 1960 that comparative studies had been undertaken in Scotland, England, Wales, Norway, Denmark, and the Netherlands. The 1965 conference brought together the in-

investigators from each of these countries in an effort to determine "the reasons for the position of the infant mortality rate in the United States."

As the introduction to the report put it:

Two features were noted about the infant mortality rate in the United States in the 1950s: (1) the virtual halt in the rate of decline of infant mortality after a long period of rapid decline, and (2) the unfavorable position of the United States infant mortality compared with that of many other countries. These two points were viewed with concern because health authorities had for many years pointed with pride to the high rate of decrease in infant mortality in the United States.

In his opening remarks, Dr. I. M. Moriyama, chief of the Office of Health Statistics Analysis of the U.S. National Center for Health Statistics, posed the problem as follows:

One of the intriguing questions is why the rate of decline of infant mortality rates for countries such as the United States, England and Wales, and Norway has been checked, whereas the rates for Denmark and the Netherlands have continued to decline without apparent interruption.

But what was more surprising was what Dr. Moriyama added:

Incidentally, the change in mortality conditions was not peculiar to the period of infancy. The rates at other ages also leveled off in the United States, but these changes came several years after the beginning of the deceleration in the infant mortality rate.

This delay in effects on the older population was, of course, consistent with the characteristic established pattern of delay in appearance of radiation-induced cancer and leukemia. And I had found that fetal and infant mortality changed within a year after the exposure took place.

An observation made by Dr. Samuel Shapiro, director of research and statistics of the Health Insurance Plan of Greater New York, seemed to lend further support to the fallout hypothesis. Dr. Shapiro remarked that the birthweight of U.S. infants had mysteriously declined since the early 1950s. This was significant, since a decline in birthweight had been well established as one of the effects of radiation on unborn children in the studies of the Hiroshima and Nagasaki survivors. And extensive laboratory animal studies had long shown that stunting of growth and reduced birthweight were produced by irradiation of the fetus, while similar effects had been observed among human infants accidentally exposed to X rays prior to birth. Furthermore, a slight reduction in birthweight, such as sometimes appears in infants whose mothers had German measles during pregnancy, was known to greatly increase the likelihood that the infants

would die in the first year of life. As it was put later in the report of the conference: "Low-birth-weight infants have a much greater chance of dying, and hence contribute significantly to neonatal mortality."

Dr. Shapiro also noted that "the decrease in the rate of decline was more serious in the non-white population." The large series of bomb tests which began in Nevada in 1951 would logically have had the greatest effects on the large Negro populations of the Southern Atlantic and Gulf states. These included the states of heaviest rainfall, over which the fallout from Nevada was blown by the prevailing eastward high-altitude winds, spreading radioactive rain on the crops that formed the staple diet of the sharecroppers and farm laborers of the south. And their children, with the poorer diet, poorer sanitary conditions, and poorer medical care prevalent among the nonwhite populations would, if they were born slightly weaker and less resistant than normal, have a greatly reduced chance of survival as compared with white children, who are generally far better off in all three respects.

According to the report, it appeared that infant mortality patterns in European countries also fitted the fallout hypothesis. According to Charlotte A. Douglas, a public health physician from Edinburgh, "In Scotland, there had been impressive declines in maternal, fetal, and infant mortality from 1935 until the early 1950s when the

decline in mortality rates became more gradual. Since then there has been only slight improvement." Thus there was a close coincidence in the times when this leveling trend began in Scotland and the U.S. Similarly, in England and Wales, as reported by Dr. Katherine M. Hirst: "The total infant mortality has declined for years, but it began to level off during the 1950s as did that of the United States." And again, just as in the U.S., there was a large increase in the number of low-birthweight babies. The same story emerged from the account of Dr. Julie E. Backer for Norway, where the clouds drifting across the North Atlantic would have rained down their fallout most heavily on the coastal mountains rising from the sea.

However, the reports on Denmark and Holland were quite different. In these countries there had been little or no leveling off in the rate of decline. A pattern was beginning to emerge. It was the more northerly countries with the heaviest rainfall that were showing the greatest effects on infant mortality. Even within England, there was a reduction in infant and fetal death rates going southward.

The high-altitude fallout clouds from the Nevada test site, carried across the Atlantic in a northeasterly direction by the prevailing jet-stream wind currents, would have deposited their radioactivity on the northern parts of Europe with much greater intensity than on the south. This pattern was confirmed by the lack of any signifi-

cant halt in the steady decline in infant mortality in France throughout the early period of Nevada testing. There, the first leveling of the decline did not occur until after the first tests in Algeria in 1960 and the large hydrogen bomb test series in the Pacific in 1961-62. These tests resulted in substantial fallout in France. And the same situation existed in Canada, which had been largely untouched by the fresh fallout from Nevada as it was blown northeastward across the U.S. and on to northern Europe.

An even more striking confirmation of the thesis that these geographical differences in mortality rates were related to fallout was contained in a study by Dr. Bernard Greenberg, a biostatistician at the University of North Carolina, whose work was summarized in the report. Since Sweden had continued to show a much smaller percentage of low-birthweight babies than the U.S., Dr. Greenberg undertook to find out if there was some cultural or genetic difference among people of Scandinavian stock that made their children less susceptible to this condition. Taking the two states of Minnesota and North Dakota, Dr. Greenberg compared the number of low-birthweight babies in counties containing the highest percentage of Scandinavians with the number in counties containing very few Scandinavians. He did indeed find a small difference of about 10 percent, the counties with more Scandinavians showing the smaller number of low-birthweight children. But this percentage was

far too small to explain the much larger difference between the U.S. and Sweden. However, he also found something he could not explain at all, namely: *all* the counties in North Dakota, regardless of the percentage of Scandinavians, showed a much higher incidence of low-birth-weight babies than did the counties in Minnesota.

This was odd. But then I remembered an article I had read many years before about fall-out in North Dakota. It was entitled "The Mandan Milk Mystery." In the back files of *Scientist and Citizen* (now called *Environment*), a magazine then published by the St. Louis Citizen's Group for Nuclear Information, I located the article, written by E. W. Pfeiffer. It described how the Mandan, North Dakota, milk sampling station operated by the AEC had, for some unknown reason, consistently shown the highest concentration of strontium-90 in milk among all the states where such measurements were taken. In fact, in May of 1963, Mandan had shown the highest levels ever recorded anywhere in the United States. The levels were consistently much higher than in nearby states, notably Minnesota.

This, then, was a direct correlation between the amount of strontium-90 in milk and the incidence of low-birthweight babies. It should therefore be possible to find large geographical differences in infant mortality based on differences in fallout levels. The various regions should show rises beginning at different times. These rises should come first in the states in the path of the

fallout from the early tests in New Mexico and Nevada—the southeastern states of heavy rainfall, along the Gulf of Mexico and the Atlantic coast from Florida to about New Jersey. These were the regions over which the high-altitude clouds generally drift from west to east, carried along by the “jet streams,” the constant 60- to 120-mile-an-hour winds blowing northeastward at altitudes of 25,000 to 40,000 feet. Only later, after the heavy hydrogen bomb testing began in the Pacific in 1953, should the high-rainfall states in the northern U.S. begin to show an upward trend. For in these tests, which continued up until the test-ban treaty of 1963, the radioactivity was deposited high in the stratosphere, beyond the influence of the prevailing winds, and so it sifted down much more uniformly around the world. But since this high-altitude debris was also eventually brought down mainly by rain and snow, then there should still be clear differences in infant mortality between the wet and dry regions of the U.S.

Searching among the reports published by the U.S. National Center for Health Statistics, I located one that had graphs of infant mortality for every state from 1935 to 1964. And indeed, the southeastern states along the Gulf and Atlantic coasts showed the first sharp leveling during the 1946–50 period following the first test in New Mexico, while the states farther to the north did not show this trend until later in the 1950s when the tests were moved northward to Nevada.

And overall, from 1935 to 1964, the least change in the steady decline occurred in the low-rainfall states of the Southwest. In New Mexico, for example, except for a brief halt in the downward trend following the first and only atmospheric test carried out in this state in 1945, the pattern was essentially the same as for France. The decline resumed and showed only a slight degree of slowing until the early 1960s, when the large amount of debris from the South Pacific tests began to raise the levels of radioactivity everywhere.

Furthermore, the lower-income groups—mainly the nonwhite population—showed the greater change, so that the effect first became apparent in this group. In fact, in some states, the decline among this group did not just slow or stop. In Arkansas, for instance, the nonwhite infant mortality rate actually began to climb again from a low point of 30 per 1000 live births in 1946 to 39 per 1000 in 1949, gradually rising still further to a high of about 42 after the peak of testing had resulted in the highest levels of radioactivity ever in the diet in 1963. In the face of generally improving living standards, such a trend was very hard to understand any other way.

In sharp contrast, the nonwhite infant mortality rates for dry Arizona and New Mexico kept right on declining. Yet the medical care and general living standards of the Indian populations in these states was not significantly better than for the black population of Arkansas. In New Mexi-

co, the rate for nonwhites in 1952 was close to 100 deaths per 1000 live births. During the ensuing decade of heavy testing, the rate declined continuously to 40 by 1961, just as steady a rate of decline as that for the white population with its much higher living standard.

But the most convincing evidence that it had to be fallout rather than ordinary chemical pollution or any genetic, cultural, medical, or economic factors came from the evidence for Hawaii. Here was an area that originally had roughly the same high infant mortality rate among nonwhites as the southwestern states of the continental United States: about 80 per 1000 births in the early 1930s. It showed a sharp downward trend in the late 1930s that brought the rate to about 28 in 1945. But then this decline suddenly halted, and the rate actually rose during 1946–1948, shortly after the Hiroshima and Nagasaki detonations and the first tests at Bikini and Eniwetok, all located directly upwind from Hawaii. The rates for both whites and nonwhites afterward resumed their decline, reaching about 21 by 1951–52. Shortly thereafter, however, they once again not only stopped their decline but actually rose to another peak of 25 between 1957 and 1960, following the large hydrogen bomb tests conducted during 1953–58 in the Pacific. Not until four years after the end of large-scale atmospheric testing in 1962 did Hawaii resume its decline, finally going below 20 in 1966, the lowest rate in its history. The hypothesis that it

was the fallout coming down with the rain fitted perfectly: The annual rainfall in Hawaii was among the highest in the world—some 100 to 200 inches as compared to less than ten for New Mexico and Arizona in the U.S. Southwest.

The worst situation in the continental U.S. existed in Mississippi, a state directly in the path of the fallout from many Nevada tests and with poor medical care and an annual rainfall of 49 inches. There, the nonwhite infant mortality rate had shown a promising decline even throughout the period of the depression, dropping from 82 per 1000 in 1930 to a low of 40 by 1946, quite close to that for the white population. But instead of declining further, the nonwhite rate leveled off in the early 1950s and then actually started to climb sharply. By 1963 it was back up to 58, an absolute increase of 45 percent in the face of a generally rising standard of living, diet, and medical care. And once again, just as in Hawaii, the rate renewed its previous decline after the cessation of atmospheric testing, reaching 53 by 1966, the last year for which data were available.

Furthermore, examination of the official fallout measurements showed that the high rainfall areas of the south and east did indeed have two to three times as much strontium-90 in their soil as the dry states of the western mountain region. By plotting the figures for twelve typical rural and urban states on a graph, I was able to ascertain that the upward deviation in in-

fant mortality was directly related to the amount of strontium-90 deposited.

The total excess infant mortality for the United States as a whole was truly enormous. Using the calculating methods employed by Dr. I. M. Moriyama, it was possible to estimate what the mortality rates would have been if the decline had not been interrupted. In the fifteen years between 1951 and 1966, the total number of infants in the United States who died in the first year of life exceeded the norm established during the previous fifteen years by 375,000. The infants were not dying to any noticeably greater degree of bone cancer and leukemia, the effects well known to be produced by strontium-90. Instead, they were dying a little more frequently of respiratory diseases, infections, and immaturity, conditions which apparently had nothing to do with the kind of gross effects everyone had been led to expect from radiation, and which would be more noticeable among those who had poorer diets and medical care.

Even more staggering were the figures indicating that for every infant that died in the first year of life there were five to ten that died prior to birth, so that the excess numbers of fetal deaths, spontaneous abortions, and stillbirths must have reached anywhere from two to three million in the United States alone.

In addition, after 1955 there had also been a sudden slowing down in the steady decline of maternal mortality. Between 1937 and 1955 the

average rate of decline in the number of women dying from complications of pregnancy and childbirth had been 12.8 percent per year. This rate of decline slowed down drastically to 2.1 percent per year between 1956 and 1962. Even more disturbing, instead of a further decline, there was actually a rise in the number of maternal deaths between 1962 and 1963, the year when fallout reached the highest levels ever recorded and fetal deaths began their sharp rise. Thus in 1963, a total of 1466 women died from complications of pregnancy and childbirth in the United States. The calculations indicated that, if the previous downward trend had continued instead of leveling off, close to a thousand of these mothers would not have died in that one year alone. For the whole world combined, the figure would have been ten times as large.

It did not take long to discover that, beginning in 1966, some three years after the test-ban treaty was signed, when only the French and Chinese continued testing in the atmosphere, the infant and fetal mortality rates all over the U.S. and Europe suddenly and quite unexpectedly began to decline once more. According to the latest Monthly Report of the U.S. Office of Vital Statistics, the drop in infant mortality that began for the U.S. after 1964, when it had reached a peak of 24.8 per 1000 births, had indeed continued into 1968, reaching 21.7, and it showed no sign of halting. Even Sweden had resumed its decline at the same rapid rate that

it had shown prior to the onset of testing. And there, the number of deaths per year had already been so far below that of the U.S. that any further decline seemed extremely unlikely. Here was clear proof that the "irreducible minimum" in infant mortality had not been reached either in Sweden or the United States.

By 1964, the year before the international conference on infant mortality had been held, seventeen other nations in the world had reached a lower level of infant mortality than the United States, according to a report published by Dr. Helen C. Chase of the U.S. Office of Health Statistics Analysis in 1967. These included: the Netherlands, 14.8; Norway, 16.4; Finland, 17.0; Iceland, 17.7; Denmark, 18.7; Switzerland, 19.0; New Zealand, 19.1; Australia, 19.1; England and Wales, 19.9; Japan, 20.4; Czechoslovakia, 21.2; Ukraine, 22.0; France, 23.3; Taiwan, 23.9; Scotland, 24.0; and Canada, 24.7. And for the United States in that year the figure had been 24.8, or 68 percent higher than for the Netherlands, when as recently as 1947 the United States actually had a lower infant mortality than that country. Among all these nations, it was the U.S. which was exposed to the most intense fallout from the tests in Nevada, and the South Pacific.

Most of these countries had been exposed to other environmental pollutants to the same or even a greater degree than the U.S. Tobacco, food additives, pesticides, and drugs were used

throughout the world since World War II. Neither could ordinary air pollution be the principal cause, since unpolluted rural states such as Hawaii and Arkansas showed far greater upward deviations in infant mortality than any of the heavily industrialized urban states in the northeast.

The conclusion was inescapable. There seemed to be no other single factor that could account for such sudden and dramatic changes on a worldwide scale. Only radioactive fallout acting mainly on the early embryo could explain these facts.

After discussing my findings with colleagues in the Federation of American Scientists, it was agreed that they would be made public at a meeting of the Pittsburgh F.A.S. chapter on October 23, 1968. The day before the meeting, I submitted two copies of the report to *Science*. Because of his interest in the subject, I also sent a copy not intended for publication to Eugene Rabinowitch, editor of the *Bulletin of the Atomic Scientists*, a journal which had long been concerned with the possible biological effects of fallout.

After delivering my paper at the F.A.S. meeting on the morning of the twenty-third, I was interviewed by Stuart Brown, a reporter from the Pittsburgh television station KDKA. A few hours after the conclusion of the interview, the phone at my office rang. It was Stuart Brown, who said that he had just called Philip Abelson,

the editor of *Science*. Brown thought I ought to know what Abelson had said.

According to Brown, Abelson had told him that my paper had been rejected by an "independent committee." This was a most unusual statement for the editor of a scientific journal to make to the press. According to long-established tenets of professional ethics, such journals are supposed to keep all editorial correspondence completely confidential. Furthermore, the paper I had just presented had been mailed to *Science* only the night before and thus could not possibly have been reviewed by an "independent committee." Abelson had obviously been referring to the revision of the Troy paper on the rise in leukemia, which contained none of the new data on fetal and infant mortality.

Next, Brown said, Abelson had gone to the files, pulled out my folder, and read statements from the supposedly confidential report of one of the reviewers that my paper was "weak in its scientific methods" and its findings were "sweeping and sensational." And at the end of the conversation, Abelson advised Brown against using any of my findings on the air.

IX

Both Young and Old

Meanwhile, I continued my search, for the investigation of infant mortality had revealed still another possible dimension of the effects from fallout. As Dr. I. M. Moriyama put it in the introduction to his 1964 report entitled "The Change in Mortality Trend in the United States":

The same kind of change in trend observed for infants appears to be taking place in the death rates for other ages.

Regarding the overall trend, he added:

The failure to experience a decline in mortality during this period is unexpected in view of the intensified attack on medical problems in the postwar years . . . there has been a growth in the volume and scope of health services in prevention, diagnosis, medical and surgical therapy, and rehabilitation, and also an improvement in their quality. The rapid growth of health insurance plans has made high quality medical care readily accessible to ever-increasing numbers of people. The rising level of living has resulted in improvement of work and home

environment, quality and variety of food, educational attainment, and facilities of recreation. Developments in medicine arising from the exigencies of a global war have become readily available for application to civilian health problems. At no time in the history of the country have conditions appeared so favorable for health progress.

In this setting, it would seem reasonable to expect further reductions in mortality. On the other hand, the possible adverse effects on mortality of radioactive fallout, air pollution, and other manmade hazards cannot be completely ignored.

In the School of Public Health library, page after page of data showed the same dramatic upward changes for chronic diseases as had occurred for childhood leukemia and fetal and infant mortality. Deaths from all types of noninfectious respiratory diseases such as lung cancer, emphysema, and bronchitis had increased especially dramatically among all age groups, as had deaths from certain other types of cancer. The overall life expectancy, particularly for adult males, had begun to level off and then actually declined again after decades of steady rise in this country and in northern Europe.

Most significant of all were the death rates due to all types of childhood cancers in the U.S. For young people of both sexes, white and nonwhite alike, there were sudden, steplike jumps in the cancer rate between 1948 and 1951. For the

white children, the rate doubled during these three years. The rate for the nonwhite children, which had held basically steady between 1930 and 1948, tripled during the same three-year period. This was to be expected if fallout was the cause, since by far the largest portion of the nation's nonwhite population lived in the regions where the fallout from the early tests came down.

When had this overall trend begun? Dr. Moriyama's report indicated that deaths from respiratory diseases and childhood cancers either declined steadily or held level throughout the 1930s and most of the 1940s, the period of rising air pollution and the tripling of cigarette consumption. Between about 1948 and 1950, three to five years after the detonation of the first bombs in New Mexico and Japan and the onset of atomic testing in the Pacific, the death rates from these diseases suddenly began to shoot up. For example, the annual death rate among white males seventy-five to eighty-four years old from respiratory diseases (not including influenza and pneumonia) was close to 110 per 100,000 per year in 1934. By 1948 it had declined to an alltime low of about 70. But after this it shot up to 190 by 1960.

The sharp upward changes in the rates for these types of chronic diseases were reflected in an overall leveling in the decline of death rates for the United States as a whole. But this effect was particularly serious in certain states, where the death rates actually rose again after decades

of steady decline. As Moriyama pointed out: "In twelve states and the District of Columbia there appears to be a marked rise in the crude death rate during the past five to ten years as represented by the trend for North Carolina." The list of the states showed that they were all either southern states to the east of New Mexico and Nevada, or states directly to the northeast of the Nevada test site. The states were Alabama, Arkansas, the District of Columbia, Louisiana, Missouri, North Carolina, South Carolina, Tennessee, and West Virginia in the South, and Nevada, Colorado, South Dakota, and Wyoming in the West. None of these states was heavily industrialized or noted for its air pollution. In fact, the opposite was true.

The number of excess deaths in the U.S. resulting from these upward changes in the death rates was calculated by Dr. Moriyama as being 300,000 during 1956-60. This was the period of heavy Nevada testing. The fallout and the excess deaths continued as testing resumed in 1961. According to Moriyama:

The estimated excess deaths is about 85,000 deaths for 1961, and 131,000 deaths for 1962.

The "excess deaths" had jumped from an average of 60,000 per year during the earlier period of testing to 85,000 in 1961, the year that the Russians detonated the largest megaton weapons ever exploded. In the following year,

when nearly 70 megatons of fission energy were detonated by the U.S. and the Soviet Union—the highest megatonnage ever exploded in a single year—the excess deaths in the United States alone reached 131,000.

The same trends were also evident in other parts of the world. A detailed report on changes in mortality trends for England and Wales by Hubert Campbell at the Welsh National School of Medicine showed that in those countries the mortality rates for the very young and the very old followed exactly the same pattern as in the U.S. Beginning about 1953–55 the total death rates for one- to four-year-olds suddenly refused to decline further. For the five-to-nine-year age groups, which the studies of Stewart and MacMahon had shown to reflect most strongly the cancer-causing effect of irradiation during early development, the mortality rate actually started to climb again. Beginning the year after the first Nevada tests, there was also the same sudden halt in the rapid decline of maternal mortality associated with complications of pregnancy and childbirth as in the U.S. These rates actually turned sharply upward in 1960–61 for the youngest group of women (fifteen to twenty-four years of age) for the first time in modern history. And there was the same sharp rise in cancer deaths of all types for the age group five to fourteen years, following some three to five years after the New Mexico test and the detonations in Japan in 1945. Both male and female death

rates jumped in a steplike fashion: from about 30 per million per year to about 60 for boys, and from 25 to about 50 for girls, all between 1948 and 1951, exactly as in the United States. That these rises could not be due to changes in statistical or classificational methods was emphasized by Campbell: "There has been no important change in the classification of these diseases during this period. . . ."

For all age groups in England and Wales, Campbell's report showed tremendous rises in leukemia for both men and women, beginning suddenly between 1947 and 1951, with the sharpest changes for the very young and the very old. Thus, whereas the leukemia death rate had remained fairly steady for men seventy-five to eighty-four years old during the fifteen years between 1931 and 1946, ranging from 50 to 80 deaths per million individuals each year, by 1954 the rate had increased to about 200. It reached 350 by 1959, an increase of about 500 percent. During the same period, the leukemia rate remained unchanged for the middle-aged group forty-five to fifty-four years old, but it increased some 50 percent for boys five to fourteen years of age. This was the rise that had prompted Dr. Stewart's study.

The data for Japan, prepared by a group of public health physicians and statisticians from the Japanese Institute of Public Health, was particularly significant, since Japan was not only exposed to the fallout from the Hiroshima and

Nagasaki bombs, but also received the radioactive debris from the U.S. Pacific and Soviet Siberian tests.

The report showed that three to five years after the fallout from Hiroshima and Nagasaki descended in 1945 the cancer rate for the ten- to fourteen-year-old children all over Japan tripled from 10 to 30 cases per million population, gradually climbing further to 40 cases by 1955 and to 50 by 1963, a fivefold increase during the period of heavy testing. For the youngest children zero to four years old, the increase was less, once again confirming the hypothesis that radiation was the causative factor as in the case of Troy. Again, the rates for the middle-aged group remained level, while the rate for those over eighty went up as elsewhere, in the case of Japan from about 3000 to 8000 per year per million individuals.

Here then was the confirmation of why the studies of the Hiroshima and Nagasaki survivors had not revealed any effects on their children. Everywhere in Japan, mortality rates had gone up due to the fallout, so that there was little or no difference between those survivors exposed to the direct flash and those who received the fallout in their diet over the years that followed.

And after the large hydrogen-bomb tests, deaths due to noninfectious lung diseases such as emphysema and bronchitis suddenly stopped declining in Japan after 1955. This was the year after the Pacific and Siberian tests filled

the air all over the world with radioactivity. In the next two years, deaths due to bronchitis, which had been dropping rapidly from 150 per million population in 1950 to a low of 40 by 1955, actually began to rise again. Thus, in the period from 1945 to 1955, when industrial growth and the accompanying smog and chemical pollution had been very great, these respiratory diseases had been declining. As in the United States, they only rose after the enormous increase of atmospheric radioactivity.

If the major factor was fallout and not the pollution produced by industry and the automobile, then Chile provided an excellent chance to test this hypothesis in a country of low industrialization, as it was the only South American country for which detailed mortality trend data was available. Since Chile was located on the west coast of South America, facing the prevailing winds from the South Pacific that release their moisture on the steeply rising slopes of the Andes, there should be upward changes in mortality following the first two series of Pacific A-bomb tests in 1946 and 1948. And these increases should be even more noticeable for the heavy series of tests beginning in 1952, involving the "dirty," uranium-clad hydrogen bombs that had produced such massive amounts of fission products.

The Chile mortality graphs instantly confirmed this prediction, especially the plot of mortality for the infants dying between the ages of one month

and one year, which showed an initial rise between 1947 and 1949 after the first Bikini and Eniwetok tests. Far more serious was the sudden and complete reversal of the overall infant mortality trend, from a steady decline to a continuous rise beginning in 1954 and persisting until 1960, the last year for which data were available.

And the same change had taken place in the total mortality rate for all ages combined. There had been a steady decline after 1933, except for small rises during the second half of the 1940s, but then between 1953 and 1955 there was a sudden and complete end to the decline for both men and women, continuing for as long as the data had been plotted.

In the words of the report's authors, a group of Chilean public health specialists:

The significance of this trend is evident if the mortality for 1960 is estimated on the basis of regression for 1933-53. The expected rate was 8.6 and the observed rate was 12.3, which means that 28,024 of the total 93,265 deaths registered in 1960 would not have taken place if the previously described trend had continued.

This did not mean that cigarettes or air pollution were not significant factors in chronic lung disease, or that heavy metals, pesticides, food additives, and other pollutants were not adversely affecting worldwide health. The phenomenon of synergism, in which combinations of two

or more biological agents have a much greater effect than one alone, is well known to modern science. For instance, it has long been known that uranium miners have ten times the normal rate of lung cancers due to breathing radioactive gas in the mines. But those who smoked died of lung cancer at one hundred times the normal rate.

However, statistics from all over the world kept indicating that radiation was the dominant factor in these worldwide changes of mortality trends. It made no difference what the social or economic system was, nor how much medical care was available, as in the very different cases of the Netherlands vs Chile. It made no difference whether infant mortality was high or low to begin with, as in Mississippi vs. Sweden. It did not matter whether there was any air pollution, or what the genetic, cultural, or dietary differences were. There was only one way to explain these worldwide, synchronous, and totally unexpected changes that did not stop at any national boundaries nor at the edges of the seas. Only the introduction of some new and enormously powerful biological agent on a worldwide scale could produce such sudden rises in death rates that could almost be termed epidemics. And this new agent clearly seemed to be the fallout that had been released into the atmosphere in quantities equivalent to tens of millions of pounds of radium, the most powerful biological poisons yet

created by man, circling the world in a matter of a few weeks and attacking mainly the weakest in every living species—the developing young and the very old.

X

The Clouds of Trinity

The geographic patterns of the changes in worldwide leukemia and infant mortality trends between 1945 and 1955 clearly matched the patterns of fallout. But so unbelievable and far-reaching a conclusion required much more evidence before the possibility of any other explanation could be ruled out. Laboratory animal experiments had shown that various pesticides, drugs, food additives, and heavy metals could apparently cause cancer and congenital defects, while air pollution and mothers' cigarette smoking were believed to be linked to fetal and infant mortality.

There was one test, however, that would effectively rule out these agents as the principal factors in the increase. It would involve the first nuclear explosion ever set off by man. Code-named Trinity, this explosion took place at dawn on July 16, 1945, in Alamogordo, New Mexico. Before this, there was no nuclear fallout in the environment, so this explosion would have to have produced a clear effect on mortality rates wherever the fallout descended.

But where had the fallout come down? There was at that time no elaborate countrywide net-

work of fallout-measuring stations. However, many eyewitness accounts of this historic explosion had subsequently been published, and, among these, a book called *Day of Trinity* by Lansing Lamont reported the event in greatest detail. Lamont, a *Time* magazine reporter, had gone to considerable lengths to trace the direction of the drifting radioactive debris, studying weather maps and conducting extensive interviews with many of the scientists involved.

According to Lamont, shortly before the final countdown at 5:10 A.M. on the morning of July 16, Dr. Kenneth Bainbridge announced to the scientists at the various observation posts that the winds close to the ground were blowing north, toward where Dr. Robert Wilson was manning an observation post 10,000 yards from ground zero. An instant after the flash of the detonation at exactly 5:29 A.M., the churning fireball detached itself from the ground and shot upward, followed by a column of radioactive dust, penetrating the overcast at 15,000 feet. The column of dust continued upward to an altitude of 40,000 feet, where it spread out in the mushroom shape that was later to become so familiar.

The lower part of the mushroom's stem was blown north toward Wilson's position, which had to be quickly evacuated. Simultaneously, Dr. Luis Alvarez and Navy captain William Parsons, flying high above the cloud cover just to the west of the test site in an observation plane, watched the head of the mushroom penetrate

the overcast and break up into three distinct sections. These sections drifted off in different directions, generally to the northeast and east. As recounted by Lamont, the largest of the three sections, a dense white mushroom trailed by a dusty-brown streamer, drifted off in a direction just slightly north of east. Meanwhile the low-altitude fallout from the stem of the mushroom cloud continued north and northeast until it covered an area about 30 miles wide and 100 miles long, gradually settling to the ground in a white mist of intense radioactivity.

By three o'clock in the afternoon, the readings on the radiation counters monitored by Alvin Graves and his wife, Elizabeth, observers assigned to the little town of Carrizozo some 40 miles just slightly north of east from the test site, started to climb rapidly. By 4:20 P.M., eleven hours after the explosion, the counters shot off scale and Alvin Graves called Dr. Stafford Warren, the chief medical officer in charge of radiation monitoring. As Lamont put it, the fate of the little town hung in the balance while the scientists and Army officers decided whether or not to evacuate it. Ultimately, they held off, and within an hour the fallout readings had dropped. Lamont reported that these were difficult hours for Dr. Warren and the officer-in-charge of the entire project, General Leslie R. Groves. "The medical dangers were most immediate of all," Lamont wrote, "but, in addition, both men knew that the

Army was not too eager to pursue too diligently the possibilities of widespread fallout."

From the fact that it had taken the fallout particles some nine to eleven hours to reach Carrizozo, it was possible to determine which portion of the cloud had gone eastward. According to the AEC publication *The Effects of Nuclear Weapons*, typical fallout particles descend at a speed of about 5000 feet per hour, while smaller particles fall more slowly. Allowing about half an hour for the cloud to travel the 40 miles to Carrizozo at the usual speed of the jet stream air currents, then, it meant that the particles that caused the Graves's radiation counters to start climbing must have taken about eight to nine hours to descend at a rate of 5000 feet per hour. Therefore, the cloud that passed over Carrizozo must have been at an altitude of between 40,000 and 45,000 feet.

Thus, it had to have been the uppermost section of the mushroom cloud that drifted just slightly north of east. And since the winds became more northerly with decreasing altitude, the lower, smaller sections would have gone increasingly northward. This estimate was confirmed by the fact that the more northerly towns were the first to receive the fallout, and those to the east were the last. The low-altitude fallout, therefore, had come down mostly in New Mexico to the northeast, while the highest portions would have been carried more nearly eastward across Texas, Oklahoma, Arkansas, and the whole

southeastern U.S., where they would be brought down mainly by the rains.

On the basis of these estimations, then, any upward changes in infant mortality should be found to some degree in the sparsely populated areas of New Mexico itself, and to a greater extent in the more heavily populated states to the east, northeast, and north. Among the more distant states generally to the east, those with the heaviest rainfall should show the largest upward changes, since the high-altitude cloud carried the smallest particles which would largely remain aloft unless brought down by rain. Furthermore, the states near the Atlantic seaboard to the northeast, such as North Carolina and Virginia, should be affected less than nearby Arkansas, Louisiana, and Mississippi, for the cloud would have gradually fanned out and the short-lived radioactivity would have steadily diminished in intensity with the passage of time.

The five-year period 1940-45 was the longest period immediately preceding the test during which a steady decline in infant mortality had existed in every state in the U.S. Thus, the amount of upward deviation from this rate of decline would provide a measure of any changes occurring after the explosion.

When the infant mortality figures were plotted on a map of the U.S., the pattern began to emerge. The states directly to the west and far to the northeast of New Mexico kept declining at the 1940-45 rate. In fact, in some cases the de-

cline was actually somewhat faster, due to the introduction of sulfa drugs and antibiotics, since the greatest cause of infant deaths was the infectious diseases which these new drugs succeeded in cutting back. But each year after 1945 and beginning strongly in 1947, there was a growing excess infant mortality in the states of the Gulf and Atlantic coasts to the east and northeast of the test site, amounting to as much as 30 to 40 percent. This pattern extended over the entire southeastern part of the country, from Texas, Arkansas, Louisiana, and Mississippi all the way across Alabama and Georgia to South Carolina, North Carolina, and Virginia, and existed both for the poorer nonwhite as well as for the socioeconomically better off white infants.

Two of my colleagues, Donald Sashin and Ronald Rocchio, became quite concerned about these findings and offered to work out a computer program to calculate and plot the infant mortality rates for every state, thus removing any possible subjective bias. The pattern that emerged from the computer was essentially the same except for one striking difference. For some reason the computer maps showed excesses in infant mortality for the north-central region of the U.S., in Montana, Idaho, Wyoming, and especially North Dakota. In 1946, even before the increased infant mortality manifested itself to the east of New Mexico, North Dakota was showing a 19 percent increase, reaching 32 percent in 1949. But the low-altitude fallout that went northward

from the Trinity test could not possibly have been significant in these distant states so directly to the north.

The mystery was solved when a colleague happened to show me a copy of an AEC publication entitled *Meteorology and Atomic Energy*. In the opening chapter, dealing with the history of the atomic energy industry, the report explained that in 1944 the first of a series of giant nuclear reactors had gone into operation in Hanford, Washington, to produce the plutonium for the Trinity bomb. The reactor was located in the dry eastern edge of the state of Washington, directly upwind from Montana, Idaho, and North Dakota. Because the operating engineers did not have sufficient experience with these enormous new reactors being built under wartime pressures, large releases of radioactive gases occurred. As the AEC report described it:

As soon as a charge of fuel came out of the plutonium production reactors, a large source of gaseous effluent was encountered. For the plutonium produced to be removed from the uranium and other fission products, it was necessary to dissolve the fuel by various chemical reactions. During the early stages of this process, all the noble-gas fission products, notably radioactive isotopes of xenon and krypton, were released. It was not feasible to remove them by a filter system; they were released to the atmosphere in rather large quantities.

Still more disturbing was the statement that "large quantities of radioactive iodine were involved." Additionally, when the reactor's fuel elements would occasionally catch fire, krypton and biologically more hazardous fission products such as strontium and cesium were driven off.

Not only could this account for the sharp rise in infant mortality in the northernmost part of the U.S. before the effects of the first bomb could make themselves felt, it could also explain the "Mandan Milk Mystery"—the inexplicably high strontium-90 content of the milk collected at Mandan, North Dakota, by the AEC's New York Health and Safety Laboratory throughout the 1950s. For the radioactive particles from the Hanford reactor in Washington would have largely passed over dry Idaho and Montana as they were blown by the prevailing westerly winds toward the wet eastern part of North Dakota where Mandan was located.

There was also one other peculiarity in the computer-printed maps. Florida, North Carolina, and Oklahoma showed no increase in infant mortality during the five years following the test even though they had been in the path of the fallout. Soon, however, when I received detailed weather maps from the U.S. Meteorological Records Center, this too fitted the hypothesis: during the week ending July 17, 1945, the heavy rains had missed these states. In fact, the weather map indicated that the rainfall for that week had been

restricted mainly to a rather narrow zone, extending from Texas along the northern edge of the Gulf of Mexico and then up the Atlantic coast. About 90 percent of the fallout is brought down by the rains, while only about 10 percent settles to the ground in dry air. This is why the zones of heaviest rainfall showed the sharpest increases. The *Weekly Weather and Crop Bulletin* summarized the rainfall situation as follows:

Rainfall during the week was again limited mostly to sections east of the Rockies. Heavy 24-hour amounts occurred the forepart of the week in Northeastern Texas and coastal areas of Gulf States and the latter part of the period in most Atlantic States and the lower Lake region.

And the summary ended: "Rain was light over the Central States and little or none occurred in the far west."

The *Weather Bulletin* also made it clear why the effects of a single small bomb had been so serious. Referring to the rainfall, it added: "Haying continued in nearly all states with generally good to excellent yields."

Thus, the fallout from Trinity, which was twenty to thirty times greater than the fallout from later similar-sized tests because the fireball touched the ground and created enormous amounts of radioactive soil and vaporized rock, was deposited on the fresh vegetables and hay being harvested that week. And so the intensely

radioactive short-lived isotopes, as well as the long-lived strontium-90, quickly found their way into the milk and food and from there into the unborn children in their mothers' wombs.

XI

The Battle for Publication

Early in January of 1969, the article on fetal and infant mortality was returned by the editor of *Science*. This was the paper that had been discredited by Abelson in his phone conversation with the reporter Stuart Brown, before the paper had arrived in the offices of *Science*. Copies of three reviews were enclosed. Two were clearly written by individuals in the field of public health and statistics; these merely contained suggestions for certain changes that might make the case more complete. The third reviewer, however, was totally negative. This individual went into a detailed analysis of fallout and dose levels in the U.S., making eight references to internal AEC reports. The majority of these reports had been prepared by the staff of the AEC's Health and Safety Laboratory in New York, directed by John Harley, the man responsible for the classified fallout measurements at Troy.

The argument used to discredit the paper in this review (and also in an article published by Harley in the *Quarterly Bulletin* of the AEC's Health and Safety Laboratory later that year) was this: According to the detailed measurements

made by the AEC's laboratory, the highest levels of strontium-90 had actually been in Utah, Montana, Wyoming, South Dakota, and Nebraska, as well as in very small portions of Massachusetts and Rhode Island. On the other hand, the lowest levels were measured in the southern U.S., from southern California to Florida. This was just the opposite of what I had said in my paper. Harley therefore argued that since infant mortality had risen most in the southern United States to the east of Nevada and New Mexico, and least in the low-rainfall mountain states of the Southwest, it clearly could not be the fallout that was responsible.

This was indeed a devastating argument, supported by a vast set of detailed measurements. Yet these measurements cited by Harley were in total disagreement with the measurements reported by the Public Health Service, which I had used in my paper. According to the Public Health Service, year after year the "wet" southern and eastern parts of the U.S. showed levels of strontium in the milk two, three, or even four times as high as in the dry western mountain states. The mystery was resolved a few months later when Dr. E. A. Martel, a former U.S. Air Force fallout specialist, told me the story of the gummed films.

The technique that had been used by Harley's lab to measure the fallout involved the use of a sheet of plastic about a foot square, coated on one side with a sticky substance very much like that used on flypaper. These plastic squares were

mounted on a stand with the gummed side facing upward so as to catch the fallout particles as they descended. Every few days the films were collected, and shipped to the laboratory, where the radioactivity was measured.

In a detailed study later carried out by scientists at the Battelle Memorial Institute in Ohio, it was discovered that in the dry states of the west, the winds constantly picked up the radioactive dust again and again, so that the exposed gummed films, acting just like the flypaper in a room full of flies, ended up collecting much more radioactivity than was typical for the soil of the area. On the other hand, in the high-rainfall areas east of the Mississippi, the rains soaked the fallout deep into the soil and kept the dust levels low. Thus, by the early 1960s, it was widely realized that the so-called "gummed film" measurements of fallout had given levels much too high for the dry mountain states, and too low for the east and south. In fact, in the 1962 United Nations Report on Atomic Radiation, of which John Harley was one of the authors, there was a note from Harley on page 225 indicating that the "gummed film" procedure "may lead to an overestimate of the tropospheric fallout," the troposphere being the lower part of the atmosphere containing the clouds of rain and radioactive dust.

Yet seven years later Harley used the "gummed film" measurements in his attempt to discredit my correlation of nuclear testing with the rises in

infant mortality, writing that "fallout before 1954 was exactly the opposite of what was stated by Dr. Sternglass."

A few days after the paper on infant mortality had been returned by *Science*, the paper on the leukemia rise in the Troy area was also returned with a note of rejection. It was the same story again. Two of the three enclosed reviews were clearly by public health physicians and statisticians and were quite favorable. One of these reviewers in fact stated that "the comparability between past fallout and past irradiated cases is 'impressive,'" while the other reviewer began with the statement: "The conclusion of this paper, if correct, is clearly a most important one." The third reviewer, however, was completely negative, and, exactly as in the case of the paper on infant mortality, there could be no doubt to which organization this third reviewer belonged. Almost word for word and point for point, the third review resembled a critique that had been sent to me a few months earlier by John Conway, Chief Counsel for the Joint Committee of Atomic Energy. And this critique had been sent to him by the AEC's Division of Biology and Medicine.

Perhaps the most remarkable point made by the third reviewer, in view of my experience with the Health Department of the state of New York, was the following: After arguing that the data on Albany-Troy was "incomplete," the reviewer asserted that "Sternglass could obtain the missing data."

And then, quite unexpectedly, less than a week after the two papers had been returned by *Science*, the following letter arrived:

Bulletin of the Atomic Scientists
A Journal of Science and Public Affairs
Eugene Rabinowitch, Editor

Dear Dr. Sternglass:

The drawings which we have for some of the figures in your article on Infant and Fetal Mortality Increase in the U.S. are not dark enough to be printed. Could you send us the original drawings—or very clear, dark copies—for figures 1 and 3.

Thank you for your assistance.

Sincerely yours,

Merry Selk
Editorial Assistant

I gradually realized that the *Bulletin* must have decided to publish the report which I had sent to Dr. Rabinowitch merely for his information.

As I later learned from the managing editor, Richard S. Lewis, in the face of strong reviewers' opinions both pro and con, it had been decided that the grave issues raised by my findings should be publicized and discussed as widely as possible, both by the scientific community and the general public. This was indeed good news. For al-

though *Science* had a far wider circulation, the *Bulletin* reached an important group of physical and political scientists in university and government circles around the world. Furthermore, like *Science*, it was carefully read behind the Iron Curtain, hundreds of copies of each issue being sent to Russia, Poland, East Germany, Czechoslovakia, and Rumania.

Subsequently, I presented all the new evidence before a meeting of the National Council of the Federation of American Scientists, and the Council voted to set up a special committee to look into the evidence in detail. Dr. John T. Edsall, a noted biologist at Harvard University, agreed to head up the study committee, and after many months of investigations during which he consulted a number of specialists, he indicated in a letter that he would personally be willing to urge the editor of *Science* to reconsider his decision not to publish my findings.

Meanwhile, at the suggestion of a mutual acquaintance, I sent copies of my data to Dr. Luis Alvarez, then president of the Physical Society, who was now heading a physics research group in Berkeley, California. I included the maps showing increases in infant mortality downwind from the Trinity test site in New Mexico, for it had been Alvarez who watched the mushroom cloud from the first atomic explosion drift off across the United States. Thus, he was one of the few individuals in the world who had firsthand knowl-

edge of the way the cloud had broken up and the directions in which it had drifted.

In the first paragraph of the reply I soon received from him, he stated that he had found the evidence "very impressive, particularly the map of the United States with the percent excess in mortality showing an effect only downwind of the Trinity site." He added that "in view of the enormous statistical significance of the results you plot on your map of the United States it is difficult to question your findings."

My article appeared in the April 1969 issue of the *Bulletin*. Interestingly, the managing editor, Richard Lewis, later told me that pressure had come both before and after publication in the form of long-distance calls from Washington from individuals who claimed to be long-term government friends of the journal. They said it was a grave mistake for the *Bulletin* to publish my article. When Lewis asked their names, they refused to identify themselves.

The April issue also carried an article by Dr. Freeman J. Dyson, a theoretical physicist at the Institute for Advanced Studies in Princeton, entitled "A Case for Missile Defense." It was clear that the basic premise of Dyson's argument in favor of an antiballistic missile (ABM) system would not be valid if my conclusions on the vulnerability of the developing infant to radiation were correct. His basic assumption was that a defensive system, once installed, regardless of

how really effective it might be, would force an attacker to concentrate many of his missiles on a few defended cities, thereby reducing the number of cities that could be attacked with a given number of missiles and saving those cities that could not be attacked. But in the process, the "saved" cities would be inundated with intensive fallout.

I wrote a short note in rebuttal to Dyson's article, hoping that it would be published in the *Bulletin*. A few weeks later, a letter arrived from the *Bulletin* containing galley proofs of my letter and a reply by Dyson that began as follows:

I welcome this chance to call attention to Ernest Sternglass' article "Infant Mortality and Nuclear Tests" in the April *Bulletin*. I urge everyone to read it. Compared with the issues Sternglass has raised, my arguments about missile defense are quite insignificant.

Sternglass displays evidence that the effect of fallout in killing babies is about a hundred times greater than has been generally supposed. The evidence is not sufficient to prove Sternglass is right. The essential point is that Sternglass may be right. The margin of uncertainty in the effects of worldwide fallout is so large that we have no justification for dismissing Sternglass' numbers as fantastic.

If Sternglass' numbers are right, as I believe they well may be, then he has a good argument against missile defense. . . .

Thus it appeared that once the evidence on the dangers of worldwide fallout was allowed to reach the scientific community at large, responsible scientists would be willing to reconsider their past judgements.

XII

Counterattack at Hanford

In May 1969, for the first time in over fifteen years, as a result of growing concern among radiobiologists, there was to be a symposium dedicated to the effects of radiation on the developing mammal, including the human infant, both prior to and immediately after birth.

Not since the detonation of the first hydrogen bombs in 1953 had such a conference been sponsored by the Atomic Energy Commission. In the ensuing years, a vast amount of data had been accumulated on the biological effects of radiation given to animals at doses comparable to those that would be expected in a nuclear war. But, even by 1969, almost nothing had been published on the more subtle effects of lower radiation doses, comparable to those from fallout, given over long periods of time. And there still had been no publication of any large-scale statistical studies of populations exposed to fallout.

It was true that the AEC had initiated a large-scale statistical study of some 20,000 persons exposed to long-term, low-level radiation in AEC laboratories. But the study population included only adults in their period of least sensitivity to

radiation, receiving the best medical care available, and working under carefully controlled and monitored conditions designed to minimize any chance of absorbing isotopes into their bodies. Since the control population consisted of individuals in the same families, communities, and installations who were not exposed to the radiation in the plant but who consumed the same fallout-contaminated food and inhaled the same fallout particles as the exposed test population, no effects from the fallout in the environment could in principle be detected by this study. More important, the study population did not include the children of the radiation laboratory workers. There was no search for unusual rises in stillbirths, infant deaths, congenital malformations, or cancer deaths among these children. When I inquired as to why this type of information was not sought, I was told that it was left out on the grounds that such questions might unduly alarm AEC employees.

There was clearly no need for the AEC to restrict its studies to so limited a population. In 1966 large statistical studies had been published regarding the very small amounts of naturally occurring radium in drinking water in the states of Iowa and Illinois. In these studies a definite increase in bone cancer had been observed for the areas with high radium levels, clearly suggesting that the effects of very low levels of long-term radiation on large populations could in fact be detected, and, furthermore, that there was

no evidence for a safe threshold even at the doses received from natural background radiation.

A letter from the Surgeon General of the United States, Dr. Jesse L. Steinfeld, to Representative William S. Moorhead of Pittsburgh made it clear that no large-scale epidemiological studies of possible low-level fallout effects were ever carried out or published by either the AEC or the U.S. Public Health Service, despite the fact that such studies had been specifically requested by congressmen Holifield and Price in the course of the 1963 congressional hearings and then promised by the Surgeon General then in office, Luther L. Terry. The exact words used by the present Surgeon General, Steinfeld, were:

. . . studies to determine the feasibility of a national program to analyze morbidity and mortality data of thyroid cancer, leukemia, and congenital malformations in relation to radiation exposure led to the decision that a national program was not indicated.

The letter added that "the feasibility studies were not published," and ended with the following statement:

Some effects on human populations from levels of radiation of the magnitude encountered during nuclear warfare or from direct high-level radiation therapy have been well established. Effects from low-level radiation have not been as clearly delineated and further research into these problems is needed. The Service welcomes

the opportunity to work with those who desire to construct and execute scientifically based plans for studies of human effects from low-level radiation exposure.

Thus, in all the years since 1963 no large-scale study of the effects of low-level radiation from *fallout* had evidently been undertaken.

The Surgeon General's letter did say that studies of fallout-exposed school children in Utah were being conducted by Dr. G. D. Carlyle Thompson of the Utah State Department of Health and that Dr. Thompson had published some preliminary data on thyroid cancers among these children in the October 1967 *American Journal of Public Health*. "Progress reports to date," wrote Surgeon General Steinfeld, "show no unusual increased incidence of leukemia and no cases of thyroid cancer among children who reside in the selected 'exposed' area of Utah-Nevada."

Examination of Thompson's article, however, showed that, although Steinfeld's statement as phrased was literally true, it was misleading in its implication. For although Thompson's paper indeed indicated no increase in thyroid cancers in these children relative to their counterparts in New York State when both sexes were combined, there was in fact a significant increase among girls zero to nineteen years of age. Among this group there was a total of ten cases for 1958-62, as compared to four cases expected. For young women aged twenty to twenty-nine

years old, the number was twenty as compared to nine expected. Still more significant, the rate of thyroid cancers per 100,000 young women under age thirty in Utah had increased almost 400 percent—from 0.6 in 1948–52, before the Nevada tests, to 2.3 in 1958–62.

Since no figures on leukemia deaths in Utah were given in Thompson's article, it was necessary to consult the U.S. Vital Statistics, and what they showed was quite unbelievable. For the age group five to fourteen years, there were large percentage rises in leukemia deaths exactly three to five years after each of the major test series that deposited fallout in the Utah area. Between 1949 and 1967, the annual number had increased fourfold in successive peaks from 1.5 to 6.2 per 100,000 children. But since leukemia rates for children five to fourteen years in New York State and elsewhere also went up when fallout became widespread, although not as much as in Utah, a statistician could perhaps say that there was "no unusual increase of leukemia" in Utah.

This, then, was the situation in May 1969 on the eve of the Hanford conference. According to the preliminary program of the symposium, Dr. Alice Stewart would be present. The conference was to take place at Hanford, Washington, the site where the plutonium for the Trinity bomb had been produced in 1944. A few days before I was scheduled to leave for Hanford, I watched the computer print out map after map of rising infant mortality stretching eastward from the

Hanford reactors and from the Trinity site at Alamogordo, New Mexico.

At about this time, a completely unexpected letter arrived by special delivery from the New York State Health Department's Bureau of Cancer Control:

April 30, 1969

Dear Dr. Sternglass:

Doctor James Yamazaki of the University of California School of Medicine has written to me about your approaching presentation at the Ninth Annual Hanford Biology Symposium. In his letter, Doctor Yamazaki requested information about environmental factors in the Albany-Schenectady-Rensselaer country areas in the early 1950s.

As you can see from my response, a copy of which is enclosed, this Bureau has reviewed some of the data upon which your June 1968 presentation in Denver was based. The results of this preliminary analysis are noted in the letter to Doctor Yamazaki. It is my opinion these data are too inaccurate for the type of analysis you have done. We are planning to restudy this subject and would be happy to have you take part.

I hope this reaches you before the May 5th conference.

Sincerely yours,

Peter Greenwald, M.D.

Director

Bureau of Cancer Control

The enclosed letter which Greenwald had written to Yamazaki indicated that J. H. Lade of the New York State Health Department had apparently made some errors in his data on leukemia in Albany-Troy, published years before in *Science* in the attempt to prove that there had been no effect from the fallout of April 1953. As Greenwald put it, "While it is unfortunate that this Health Department may have erred in not clearly describing the possible inaccuracies in this table it is clear that they should now be pointed out." Upon examining Greenwald's revised table, I discovered a remarkable coincidence: These newly found inaccuracies concerned only the cases born in 1953, the year the fallout arrived. Greenwald's new figures showed a greatly reduced number of leukemia cases in that year, thus making it now possible to argue that there had been no significant effect from the fallout.

At the end of Greenwald's letter was the notation:

cc: Dr. Sternglass
Dr. Sagan

And on the preliminary program of the Hanford meeting, listed below my name, was:

9:40 P.M. INVITED DISCUSSANT
L. A. Sagan
Palo Alto Medical Clinic

This was Dr. Leonard Sagan, who had long been working for the AEC's Division of Biology and Medicine. He was apparently scheduled to be discussant, or critic, for my paper at Hanford. Why was he listed as being affiliated only with the Palo Alto Medical Clinic? Interestingly, my paper was the only one in that session for which a discussant had been listed in the final program. The co-chairman of the session was Dr. J. N. Yamazaki of the University of California, the same individual who had just requested information on the Troy rainout from the New York State Health Department. Dr. Yamazaki had published a number of papers while working for the Atomic Bomb Casualty Commission that argued for the absence of significant leukemia rises among offspring born to the survivors of Hiroshima.

Upon arriving at Hanford, I was interested to discover the summary of a paper written by Dr. Y. I. Moskalev and his colleagues at the Institute for Biophysics in Moscow. This group of Russian scientists had studied the effects of low-level radiation—of the type produced by fallout—from strontium-90 and other isotopes when given to various animals before and during pregnancy. Their observations on the offspring of all types of animals, ranging in size from rats, rabbits and dogs, to sheep and cows, appeared to show similar effects as I had postulated for human infants. There was no spectacular increase in gross malformations at low levels of strontium-90. Instead there was a small reduction in weight at birth, a

decline in fertility, and an increase in the number of fetal deaths. And the earlier the radioactivity was fed to the pregnant animals, the more pronounced were the effects, just exactly as the studies of diagnostic X rays given during pregnancy had indicated in the case of man.

Unexpected difficulties, however, developed, and the Russian scientists were unable to attend. This was a great disappointment, for the studies of Dr. Moskalev and his group were the first published work I was aware of which appeared to offer the kind of crucial laboratory confirmation of the statistical evidence for the damaging effects of fallout on the newborn.

However, early in the conference, other evidence was presented in session after session that internal radiation from isotopes, of the type which occurred in fallout was particularly hazardous to the ova and the early embryo and fetus. These were levels much lower than those that led to lethal effects in mature animals.

First on the program the evening my paper was scheduled was a paper by Dr. M. L. Griem of the University of Chicago School of Medicine, Dr. Griem indicated that in his careful study of children X-rayed while in the womb there was no evidence for a difference in the number of leukemia cases between the irradiated children and the two control groups who had not received any radiation. But as soon as Dr. Griem had begun his talk, it became apparent that the number of children X-rayed was only 1008, a group

in which one would normally not expect more than one case of leukemia in ten years of life. Thus, even a doubling would at most result in one extra leukemia case, clearly too small an effect to be readily observed with any statistical certainty. As Dr. Alice Stewart herself pointed out in the ensuing discussion, "the reality of juvenile leukemia is such that no one could hope to do a follow-up like this and detect an increase of 50 percent of the normal incidence without being prepared to follow out 900,000 children for ten years, and no one has done that." She herself had to interview the families of over 7000 children who had died of cancer out of a total of nearly nineteen million children born in order to establish a clear causal connection.

Interestingly, however, Dr. Griem's study actually lent support to the hypothesis of other effects from low-dose radiation, even in so small a study population as 1008 children. His data showed that, although there was no detectable increase in leukemia, there was a significant increase in benign tumors and certain types of congenital birth defects, especially the severely disfiguring birthmarks. Furthermore, Griem's study revealed that stillbirths had nearly doubled—from eight and nine cases in the two control groups to sixteen among those who had received X rays in the womb. And according to Griem, the X rays were given mainly in the less sensitive second and third trimesters of pregnancy, with doses of only 1 to 3 rads to the fetuses. But Dr.

Stewart had found the sensitivity to be some ten times greater in the first three months of pregnancy. Thus, Griem's study indicated that if the X rays had instead been given in the earliest developmental phase, a dose only about one-tenth as large, or a mere 100 to 300 millirads, would have produced a doubling of stillbirths and genetic defects. These were in fact the general doses that had been received by large numbers of unborn children from fallout. So Griem's study actually provided, for the first time, direct observational evidence in man for my statistical findings on fetal mortality.

Soon it was Dr. Stewart's turn. As her opening paragraph made clear, it was the opposition to her findings that drove her to find more and more convincing evidence:

More than ten years ago, a retrospective approach to the problem of cancer etiology led to the conclusion that about one in twenty of the 600 cancer deaths that were taking yearly tolls of the seven million children living in Britain were being caused by obstetric X-ray examinations. So far as the investigators themselves were concerned, the matter would probably have rested there had it not been for the general skepticism that greeted this suggestion.

However, so many clinicians and experimentalists refused to accept the necessarily epidemiological evidence that it was finally decided to remove all grounds for doubt by isolating groups of mainly radiogenic cases (i.e. X-rayed cases)

and wholly non-radiogenic cases (i.e. non X-rayed cases) and observing the ages of the two groups of children when symptoms developed.

She then went on to present new evidence that leukemia cases among children X-rayed while in the womb showed a clearly recognizable, narrow age range of between three and five years after birth, evidence that supported the hypothesis that the excess cases in Albany-Troy among the children irradiated in the womb were caused by radiation, since they too showed the same abnormal pattern of age at death.

Dr. Stewart also observed that all types of cellular defects produced by radiation "were felt with equal force by all systems and organs" in the fetal stage, a point crucial in explaining how radiation could cause subtle types of damage that would lead to such biochemical defects as the inability to resist infections early in infancy or childhood. And in her conclusion, she pointed out that her evidence clearly showed that the effects which prove fatal before the age of ten years are mainly initiated before birth, and often in the most sensitive period of early development. Thus Dr. Stewart's paper had greatly strengthened my statistical findings on the effects of fallout radiation.

I then presented my evidence, concluding with the estimate that some 400,000 infants of less than one year of age probably had died as the result of nuclear fallout between 1950 and 1965.

The chairman then announced that there would be no discussion until Dr. Sagan of the Palo Alto Medical Clinic had commented on the paper. Dr. Sagan took the microphone.

He began by showing the slides of the material collected for him by the cochairman of the session, Dr. Yamazaki, from the New York State Health Department. Sagan made the point that the data was too unreliable for the type of study I had done. He next cited an estimated figure for the expected number of leukemia cases in a population exposed to a radiation dose of 1 rad, namely one case per year among each million persons. Therefore, he concluded, the doses in the Troy area, with its 130,000 children, were some 1000 times too small to produce even a single extra leukemia case. Sagan's estimated figure, however, was the one for the mature adult, who is least sensitive of all to radiation. It was evident from Dr. Stewart's and Dr. MacMahon's work, as well as from the paper just delivered by Griem, that the fetus and especially the early embryo were many hundreds of times as sensitive as the adult. Furthermore, the total dose from internal radiation in Troy was at least five times the external dose measured by the AEC's Health and Safety Laboratory—which Sagan had used in his argument.

He next cited Harley's invalidated "gummed film" fallout measurements to discredit any association of fallout with the infant mortality rises in the wet areas of the South and East. He did

this despite the fact that the Public Health Service's data for strontium in the milk that I had just presented correlated perfectly with the changes in infant mortality in every state for which the Service had published data.

The discussion was then opened to the floor, and a long and often heated debate began. Some AEC scientists had slides which showed that miniature swine fed fairly large doses of strontium did not seem to show detectable reductions in weight and litter size. This was in apparent contradiction to the findings of Moskalev, who was unfortunately not present to defend his work. But then Dr. T. K. Ellis of the University of Utah reported that in beagle dogs given small amounts of strontium-90, surprisingly strong effects on male hormone production had been found, as well as changes in reproductive cells. Dr. Harold Rosenthal, who had made measurements on baby teeth collected from all over the United States, indicated that for Texas, St. Louis, Toronto, Detroit, Chicago, and California he had found a close correlation between levels of strontium in the teeth and the levels reported for the milk. This showed that the Public Health Service's milk measurements which I used in my work were indeed a good indication of the amount of exposure received by infants in various areas.

The evidence supporting unexpectedly severe effects on the early embryo and fetus from internal radiation sources continued to accumulate

until the close of the conference. And afterward, when I was able to examine the original manuscript submitted by Dr. Moskalev, I found that his studies did in fact confirm the most crucial points at issue in the whole fallout controversy. His group had found that when various of the isotopes contained in fallout were fed to female animals during pregnancy large fractions were transferred through the placenta to the developing fetus. For example, up to 38 percent was transferred in the case of strontium given to rats and up to 66 percent in the case of cesium given to a litter of dogs. Furthermore, the amounts reaching the developing fetus were many times greater for continuous, slow intake (such as occurs with fallout in food) than for a large single dose.

More significant, Moskalev had found a direct relationship between the size of the doses of isotopes given just before pregnancy and the percent of the offspring that died—even for doses as small as 4 billionths of a curie per gram of body weight. This dose was well within the range of doses from fallout delivered to the early human fetus by the accumulated strontium in the mothers' bones. Thus, the argument that there might not be any effects at all from long-term, low-level radiation as opposed to doses given all at once, like diagnostic X rays, had now been disproven by direct experiments.

Moskalev's results also showed that, regardless of whether a given amount of isotopes was

fed to a mouse weighing 20 grams or to a dog weighing many thousands of grams, approximately the same fraction of the total amount would always concentrate in the rapidly growing embryo. This was one reason why, in all species, the fetus was so much more sensitive to fallout radiation than the adult. A given tiny amount of an isotope in the body of an adult might be quite tolerable if it was evenly distributed throughout the 70,000 grams of an average woman's body weight. But if even only one one-hundredth of this amount in the mother's body goes to the embryo during the first two to four weeks of development, when the embryo weighs less than a hundredth of a gram, then the concentration of radioactivity in the embryonic tissue would be the same as if the entire original amount had been given to a 1 gram embryo. Instead of the relatively minor radiation dose produced by this original amount distributed in the 70,000 grams of body weight typical of the adult, there would be a concentration 70,000 times greater in the early embryo. And added to this was the fact that the embryo was already hundreds of times more likely to develop cancer or other forms of biochemical damage than the much more resistant, fully mature adult, as paper after paper presented at the conference showed. Thus, the low external doses given by fallout to the body of an adult, doses on which the world's radiation protection agencies had based their assumption that fallout was harmless,

were actually highly lethal doses for the early embryo.

It was not our physics and technology that had been inadequate, but our knowledge of biological systems and their enormous ability to concentrate toxic agents. Just as in the case of DDT, it was not the amount distributed throughout the environment that was so serious. It was the selective concentration in the food chain and then in the newly forming organs of the rapidly developing young embryo. Since all higher animals, including man, must pass through this critically sensitive phase, it was clear that, unless the problem was widely recognized and acted upon, man could extinguish himself and all other animals, not through the effect of radiation on the adult, but through the effect on the weakest link in the chain of life—the unborn and the very young.

XIII

The Public's Right to Know

Not a word of the findings reported at Hanford appeared in the national press or the other public news media. No science writers had been present, and none of the wire services had covered the meeting, held in a remote part of the country. Yet in view of the accumulated evidence presented at Hanford, it was now clear that the sensitivity of the early embryo was so great that if a nuclear war ever broke out only the more resistant reptiles and insects would survive the lingering radiation.

In June 1969 I wrote a letter to *The New York Times* describing the Hanford findings and their implications for the ABM debate. Shortly after the letter appeared, I received a phone call from the New York correspondent of *The London Observer*, Joyce Egginton. The first thing Miss Egginton asked was why there had not been any stories about these startling findings either in the news section of *The New York Times* or in any other newspapers served by the national wire services. In particular, she wondered why nothing had been reported about my findings in view of the fact that I had just presented them again at a

recent meeting of the Health Physics Society. There, the University of Pittsburgh's public relations department had made arrangements with the public relations department of the Health Physics Society for an interview with an individual who stated that he wanted to write a story on my work for the Associated Press. After the presentation of my paper, this individual introduced himself, stating that he was in charge of the newsroom for the Society but also often wrote stories for the Associated Press. He already had much of the story written and merely wanted to check certain details. It seemed to be quite a good account. But his story was never carried by the Associated Press. Not until later did it become clear what had happened.

Meanwhile, the editor of *Esquire*, Harold Hayes, called soon after Joyce Egginton to ask whether I would be willing to write a story on my findings for his magazine. In view of the fact that the general public had heard nothing at all of the findings presented in Pittsburgh or at Hanford, I accepted.

A week or two later, I received a call from Hal Stromholt, a writer for the Associated Press office in Pittsburgh. He said that the AP had asked him to do a story on my findings presented at the recent Health Physics Society meeting, which Joyce Egginton had just reported in *The London Observer*. I asked him why the AP had not carried the story that had been written for it many weeks before. Surprised, Stromholt asked

the name of the man who had done this earlier story. I told him, and he then said that no one by that name either worked in the AP office in Pittsburgh or was employed as an occasional writer or "stringer." Furthermore, he added that no earlier story could have been sent to the AP, for they certainly would not have asked him to write a second one on exactly the same news item. Significantly, the Associated Press did in fact use Stromholt's story, and it was carried throughout the country.

Shortly thereafter the publishers of *Esquire* decided to stop the press run on the September issue in order to include the article on infant mortality and nuclear testing as a special insert. Furthermore, in view of the serious implications for the decision on whether the U.S. should build the new antiballistic missile nuclear defense system, which would fill the atmosphere with hundreds of times as much fallout as all of the past nuclear tests if it were ever to be used, the publishers had decided to take out full-page advertisements in *The New York Times* and *The Washington Post* that would summarize the principal points of the article. They felt that these advertisements might appear in time to be considered in the Senate debate. Harold Hayes, the editor, also sent advance copies of the article to every congressman and senator, together with a personal letter explaining the reason for this unprecedented action on the part of his magazine.

But for the ABM debate, it was too late. The final vote came only eight days later with a narrow victory for the Defense Department and the AEC, before the evidence of biological risks had a chance to be fully considered by Congress and the public.

On October 12, 1969, a few weeks after the *Esquire* article appeared, *The New York Times* published an account of a critique of my Troy data prepared by Dr. Peter Greenwald, director of the New York State Health Department's Bureau of Cancer Control, and Mrs. Sandra Kinch, director of the Department's Office of Biostatistics. According to the *Times*, these two officials said that their analyses "tend to refute the validity and the conclusions" of my studies, which they described as having "definite factual errors" in the data used. To prove this, they submitted their own table of leukemia deaths among children under age ten in the Albany-Troy-Schenectady area. The figures they gave were as follows:

<i>Birth Years</i>	<i>No. of Deaths</i>
1940-42	6
1943-45	8
1946-48	9
1949-51	7
1952-54	10
1955-57	13
Total:	53

But the figures supplied by Dr. J. H. Lade in his 1964 letter to *Science*, which I had used in my study, were (when regrouped in similar three-year periods):

<i>Birth Years</i>	<i>No. of Deaths</i>
1943-45	9
1946-48	8
1949-51	9
1952-54	15
1955-57	13
Total:	54

This was indeed peculiar. Five cases were missing from Greenwald's table in the critical 1952-54 period, yet the overall total number of cases was nearly the same in both tables. Examination of Greenwald's table showed that he had added a new category not present in Lade's: the 1940-42 period, for which he listed six cases. He had also deleted two cases from the 1949-51 period, with the final result that the total number of cases was nearly the same for both tables—even though the exact five cases which indicated a large increase in leukemia during the critical years 1952-54 were missing.

Why had these five cases been removed? I re-examined the correspondence I had received from Greenwald. In it he stated that inaccuracies had been found in Lade's table among five of the eight cases born in the year 1953. But should these five cases have been eliminated from the 1952-54

entry? The answer could be found by examining Greenwald's critique, in which he described the nature of the inaccuracies in these cases. He stated that two of the children had actually been born in 1952 instead of 1953, while another had been born in 1954. But in his table in the *Times*, Greenwald had used the time period 1952-54, *not* 1953. Therefore these cases should have been included in his table by his own criteria. The fourth missing case was born in Montgomery County, New York, according to Greenwald, and not in the Albany-Troy area. But Montgomery County is right next to Albany-Troy, so this case could certainly have been caused by the fallout. And the same would hold true for the fifth and last child, who was born in New Mexico, according to Greenwald, and brought to Troy as an infant, and thus also would have been exposed to the fallout in the diet. I had hypothesized in my study that the fallout could have caused leukemia by genetic damage to the parents before conception or by direct damage either to children in the womb or to young infants. Therefore, there was no reason to exclude *any* of the five cases from the evidence. If they had been included, Greenwald's table would have shown fifteen cases for 1952-54, a doubling over the average rate for the previous years. Furthermore, the thirteen cases in the 1955-57 period were consistent with the hypothesis of genetically-caused leukemia.

Other distortions of data were present in the critique by Dr. Greenwald and Mrs. Kinch. But

how would the general public ever suspect? This was the voice of the New York State Health Department and not the AEC. Readers of the nation's press would simply assume that responsible and independent public health officials had proven that fallout was harmless after all.

During this period, amid the resultant publicity surrounding the article in the *Observer*, I was invited to appear on the NBC-TV "Today" show. According to Hugh Downs, one of the reporters on the program, the AEC had learned of the plans for my appearance and called the producer, urging him not to invite me. When the producer refused, the AEC urged that a scientist holding an opposing view should be present to give an immediate rebuttal. When this also was turned down, the AEC insisted on equal time as soon as possible after my appearance so that two independent scientists would be able to present the argument against my thesis. NBC finally agreed to provide equal time the following week.

On the program itself, Hugh Downs brought this out into the open: "I was just going to say that the Atomic Energy Commission called us yesterday. They were concerned about your appearance on the program today."

I asked the identity of the independent scientists whom the AEC wanted to present. Downs replied that one was a physician by the name of Dr. Leonard Sagan from the Palo Alto Medical Clinic, and the other was a Dr. John Storer from the Oak Ridge National Laboratory. Both, of

course, had worked until very recently for the Atomic Energy Commission's Division of Biology and Medicine. As to the reason why the AEC had urged NBC to cancel my appearance, Hugh Downs reported: "They say that their experiments with animals show that there is no damage to offspring at all from parent animals given strontium-90 in those low dosages that we get." This was in complete contradiction with the evidence submitted at Hanford by Moskalev and numerous other independent scientists. The explanation lay, as usual, in the manner in which the AEC studies were constructed.

This point was to be confirmed a few months later in a most unexpected manner by a chief scientist at one of the AEC's own laboratories. I had received an invitation to present a paper on my other research work regarding the reduction of diagnostic X-ray doses at a meeting in San Francisco in October 1969. I was also asked whether I would be willing to debate my fallout thesis on the Berkeley campus with Dr. Arthur Tamplin, who had written a critique of my work that was about to be published in the *Bulletin of the Atomic Scientists*, and to which I had just written a reply. Tamplin was a biophysicist at the nearby Livermore Laboratory, operated for the AEC by the University of California. During the question period following our debate, someone in the audience brought up the argument that AEC studies had found no significant increase in mortality among the offspring of experimental

animals fed strontium-90 for long periods of time. Immediately, someone else in the audience stood up and asked to comment on this question. It turned out to be Dr. John Gofman, Tamplin's supervisor, who was Director of the Biomedical Division of Livermore and an Associate Director of the laboratory. This was the man the AEC had placed in charge of all their radiobiological studies at Livermore back in 1963, when the hazard from internal fallout doses first aroused widespread public concern. For years Gofman had been studying the possible connection between radiation, chromosome defects, and cancer. He told the audience that he had investigated all the animal experiments carried out by the AEC, and in no case had they been designed to detect the kind of small reduction in birth-weight and ability to fight infections that I had suggested as the likely cause for the increased infant mortality in man. He concluded that, to the best of his knowledge, there was not a single animal experiment that would contradict my hypothesis, and with that he sat down.

Within less than a year, both Gofman and Tamplin publicly denounced as complete falsehood the position of the AEC as expressed by Sagan and Storer on the "Today" program, namely, that "the levels of radiation to which the American public was exposed from fallout have been harmless." As told by the two scientists themselves in the pages of *The Bulletin of the Atomic Scientists*, their public denunciation of

the AEC's position on low-level radiation effects was precipitated by the attempt of the AEC's top management to force Tamplin to suppress his own independent calculations, made in his original critique of my findings, that perhaps as many as 8000 infant deaths per year might have taken place as a result of genetic damage from nuclear testing. Dr. Spofford English, the Assistant General Manager in charge of the AEC's entire research program, together with the head of the Division of Biology and Medicine, Dr. John Totter, as well as Dr. Leonard Sagan and Dr. John B. Storer, had indicated to Gofman that Tamplin should publish his critique minus his own estimate of the possible fetal and infant deaths, thus effectively keeping this information from the public.

As Gofman and Tamplin put it: "They wanted us by *omission* to support their incredible position as stated on the "Today" show, and to put Tamplin's estimate into a less widely read scholarly journal, where it would evidently not be seen by the scientific community at large, the general public, and their elected representatives in Congress."

Subsequently, Dr. Gofman resigned his position as Associate Director of the Livermore Laboratories, and all but one of Tamplin's research group of twelve people were taken away from him six months later. Both scientists have continued to testify before various congressional committees that there is no safe threshold of

radiation exposure and that presently permitted radiation exposure levels must be cut back to virtually zero. They have proposed that no release whatsoever of radioactive materials into the environment should be permitted without a full, nonpartial, interdisciplinary examination of each situation. And they have assembled a vast body of data indicating that if the radiation doses now allowed by AEC regulations (an average of 170 millirads per year for the entire population and not more than 500 for any single individual) were to be received by the entire U.S. population as a result of peacetime uses of nuclear energy, there would be at least 32,000 and perhaps as many as 64,000 additional deaths each year from cancer and leukemia alone. And these figures did not even include fetal and infant mortality or any more subtle long-range effects on health.

At the hearings on the environmental effects of electric power generation held by the Joint Committee on Atomic Energy in November 1969 and January 1970, Gofman and Tamplin presented their conclusion that a direct relationship exists between low-level radiation doses and the development of cancer, not only in the fetus and infant, but also in the mature adult. Furthermore, they urged an immediate tenfold reduction in the permissible radiation doses to the general population from all peaceful uses of nuclear energy.

XIV

The Price of Secrecy

As late as the spring of 1970, I believed that the radiation resulting from the normal operation of nuclear power plants was so low as to present no significant hazard to public health. This belief was based on the results of an old study of emissions from the first commercial nuclear electric power plant, located at Shippingport, Pennsylvania. Conducted almost ten years earlier, it had measured the radioactivity in the cooling water taken from the Ohio River both before and after it had passed through the plant. It found that the plant added so little radioactivity to the water that there were times when the chemically filtered water leaving the plant was actually less radioactive than the river water entering the plant—especially during periods of heavy fallout.

It therefore seemed reasonable to expect that if such low levels of radioactive waste releases had been achieved in 1957 in the very first nuclear power reactor built in this country, then the later, more advanced plants would release even less. But early in 1970 I discovered that this was not the case. In the published record of the hearings on the environmental effects of electric power

generation, held by the Joint Committee on Atomic Energy in November 1969, there were tables supplied by the AEC listing the amounts of radioactivity discharged into the water and air by commercial nuclear power plants in the United States. Many plants were listed as actually releasing hundreds of thousands of times as much radioactivity into the air as others. For example, in 1967 two reactors had discharged as much as 700,000 curies, while another had released only 2.4 curies, or some 300,000 times less.

These were truly enormous quantities. Some of the many different isotopes contained in these gaseous and liquid discharges, such as cesium and strontium, were regarded as hazardous at levels as low as one ten-billionth of a curie per day in milk or food. A single curie of iodine-131 could make 10 billion quarts of milk unfit for continuous consumption according even to the existing guidelines adopted by the federal government. Such large releases of radioactivity were in fact comparable to fallout from small tactical nuclear weapons. Although dilution in the air would reduce the hazard to people living more than fifty miles away from these plants, those living nearby were unknowingly accepting vastly greater risks to the health of their children.

Furthermore, the *permissible* levels listed for many of the reactors were enormous. For the Dresden reactor, located some fifty miles from Chicago, which had emitted 260,000 curies of

radioactive gases in 1967, the permissible amount had been set at 22,000,000 curies per year by the AEC. Thus, in terms of permissible levels, the huge amount actually released could be, and was, cited by the power company as representing only about 1 percent of the maximum levels allowed.

Curiously, there was no listing given in the record of the hearings for the Shippingport plant. Why had the AEC left it out of the material presented to Congress? A few months later, at a meeting of the Health Physics Society, Charles Weaver, Director of the Division of Environmental Radiation of the Public Health Service's Bureau of Radiological Health, presented the results of studies just published in March of 1970 on the radioactivity emitted by a series of nuclear plants. According to his report, in 1968 Shippingport had emitted a grand total of only 0.001 curies into the air, or 240 million times less than was released the same year by the Dresden reactor near Chicago, Illinois. And the same report also explained why some reactors released so much more radioactivity than others. They were grouped according to type of design, and it was only the boiling-water type reactors that showed such very large releases, while the pressurized-water reactors, which included Shippingport, consistently showed the lowest waste discharges. Some of the commercial pressurized reactors did, however, emit much more than others, and all of them discharged significant

quantities of radioactive tritium into the cooling water which was then released into the surrounding rivers and lakes.

Why weren't all the reactors designed like Shippingport, so as to release the smallest amounts of radioactivity? The answer could be found in the history of reactor development. The pressurized-water reactors, like Shippingport, were originally designed for use in nuclear submarines by Westinghouse under the direction of Admiral Hyman C. Rickover. Since they had to operate for long periods in a sealed, submerged vessel, these reactors had to be designed with a minimum of radioactive leakage either into the submarine, where the crew had to live for months at a time, or into the water, where the bubbles of radioactive gases would permit easy detection of the submarine's position. The Shippingport reactor was in fact a prototype naval propulsion plant owned by the Navy and the AEC, and not a commercial power plant at all.

Meanwhile, the General Electric Company was encouraged by the AEC to quickly develop a new type of large power reactor that would be cheap and efficient enough to compete successfully with the fossil-fuel-burning electric power plants in widespread use. For the more complex pressurized reactor with its double cooling loop, although safer, was too expensive. And so GE developed the much simpler boiling-water reactor. This design, in which economic considerations were the major factor, sacrificed pro-

tection against radioactive leaks in favor of lower cost and greater efficiency of operation. Experiments showed that corrosion was a more serious problem in the single-coolant-loop GE design. Large amounts of fission products would inevitably build up rapidly in the coolant and escape through pipe joints, valve packings, and high-speed rotating shaft-seals to be discharged into the air and water. Thus, if a cheap, economical way to generate large quantities of electric power was to be demonstrated quickly so as to convince the utilities to go nuclear, there was only one solution: set the permissible amounts of radioactive waste discharges into the environment so high that the actual releases would always be well below this limit.

By 1959, the first large boiling-water reactor plant was completed at Dresden, Illinois, and in August of 1960, the first electricity from the 200-megawatt Dresden generators began to flow into the power grid of the Commonwealth Edison Company, serving the people of Chicago. The releases of radioactive gases into the atmosphere were relatively low in the first full year of operation, and so were the discharges of tritium, strontium-90, and other isotopes into the Illinois River. In 1961, only 0.158 percent of the maximum allowable amount had to be released into the air, and even liquid wastes were held down to 6.3 percent of permitted levels. Compared to the amounts of radioactivity then being released into the water and the air by the renewed test-

ing of nuclear weapons, this was certainly quite small.

But signs of trouble began to appear the very next year. By the end of 1962, corrosion had begun, and the amount of radioactive gas that had to be discharged into the air increased by almost ten times to 284,000 curies. Even the radioactivity discharged into the river rose more than three times. By 1963 emission of radioactive gases had been successfully brought back down to 71,600 curies by the replacement of leaking fuel rods, but the corrosion continued and gaseous releases shot up to 521,000 curies in 1964.

No longer were the radiation doses to the surrounding population negligibly small compared to background radiation, as everyone had hoped. Annual average external doses to the population within a few miles of the plant could be estimated at 20 to 30 millirads by 1964. This was fast approaching the 88 millirads that the people in the area normally received from cosmic radiation and natural radioactivity in the soil and it compared with what had been produced by weapons fallout.

It was becoming clear that the permissible levels of radiation from nuclear plants could not be lowered, as some scientists were beginning to urge, without having to shut the huge plant down only a few years after it had been built at a cost of well over a hundred million dollars. In fact, pressures were actually building up from

industry and the military to *raise* the permissible discharges to the environment from nuclear activities, especially in the event of an accidental heavy release from a reactor or from fallout if weapons tests in the atmosphere were ever renewed. And so, in 1964 and 1965, the director of the Federal Radiation Council, Dr. Paul C. Tompkins, who had previously served as Deputy Director of the AEC's Office of Radiation Standards and Director of Research in the Bureau of Radiological Health of the U.S. Public Health Service, announced a twentyfold rise in the permissible amounts of the most hazardous isotopes in milk in the event of an accidental release. For the first time in the history of radiation standards the permissible doses to the public were raised rather than lowered, despite the mounting evidence that there was no safe threshold dose of radiation as presented in August 1963 before the Joint Committee. And this was done quietly by presidential executive order, for which no public hearing is required.

When in 1966 the gaseous discharges from the Dresden plant had climbed to 736,000 curies, or more than twenty times what they had been in 1961 and more than twenty million times more than Shippingport had released the same year, a decision was made to start replacing the corroding stainless-steel-jacketed fuel rods with more resistant, but also more expensive, zircalloy-clad fuel. By this time, the liquid releases, containing iodine, strontium, cesium, and other highly toxic

elements, had risen to forty-three times their initial value, and, instead of being a small fraction of the permissible level, they had actually reached a full third of the AEC standards. Enormous quantities of these isotopes went into the Illinois River, flowing past Peoria, where the river water began to be used for drinking, and on to the Gulf of Mexico, concentrating thousands of times higher in the fish and in the birds that fed on them.

The example of Dresden clearly showed that it would not be possible to lower permissible radiation levels without having to shut down the whole series of boiling-water reactors that had now gone into operation all over the United States, each having cost some one hundred million dollars. And construction would have to be halted on dozens of even larger reactors in various stages of development throughout the United States and the rest of the world.

On June 6, 1970, just a few months after all the new reactor emissions data had been published, the British medical journal *Lancet* printed a full account of Dr. Stewart's fifteen-year study of the increase in leukemia and cancer among the nineteen million children in England and Wales that were born between 1943 and 1965. Her conclusions were now statistically unassailable: doubling the number of X-ray pictures doubled the risk of leukemia and other cancers, and there was no evidence for a safe threshold even at a single diagnostic X ray. One modern pelvic

X ray gives about the same dose to the fetus as is permitted for the general population by the existing federal radiation standards (175 millirads). And when the radiation was given in the first three months of pregnancy, Dr. Stewart's data showed that a mere 80 millirads—about the dose that was received from external radiation alone by the people living near the Dresden reactor in the peak year of emission—would double the spontaneous rate of leukemia and cancer in the children before they reached the age of ten. And there had been significant rises in leukemia and fetal and infant mortality in the Troy area at similar external dose levels of only 50 to 100 millirads.

There was thus little doubt that detectable health effects should have occurred in the areas surrounding the Dresden plant and other reactors. But it would require a considerable effort to collect the data from the volumes of the U.S. Vital Statistics, and I had no one to help in such a task, for I had been unable to obtain any funds for such studies. Fortunately, a group of students, who had become interested in environmental pollution as a result of Earth Day, indicated a willingness to help. The group divided itself into small teams and each took on the task of gathering the data for a particular nuclear installation.

Since most of the fission products emitted by reactors are short-lived, persisting only for anywhere from a few days to a few months, it ap-

peared that the effects on infant mortality would be sharp and immediate, just as had occurred with the short-lived isotopes in the case of fall-out. There would probably be no significant residual effect, and so the rises and falls in infant mortality should correlate closely with the rises and falls in the reactor releases.

In October 1970, we examined the infant mortality rates in the counties around the Dresden reactor. In 1966, within a year after the emissions rose sharply from the relatively low value of 71,600 curies in 1963 to 610,000 curies in 1965, the infant mortality rate in Grundy County, where the reactor was located, and in adjacent Livingston County, jumped by 140 percent, or to more than twice its 1964 value. While only thirteen infants in these two counties had died in the year after the minimum radioactive emission, by 1966 this number had jumped to thirty. And the number of babies born live in these two counties actually decreased slightly from 1170 to 1082 in 1966, so that the jump in rates per 1,000 births was actually even larger.

There could be little doubt about the statistical significance. Established statistical estimation techniques showed that the possibility of such a fluctuation being accidental was much less than one in 10,000. But this was not all. The students had gathered the data for all five counties surrounding Grundy County, as well as for a control group of six counties as far to the west and north of Grundy as possible within the state of Illinois,

counties that bordered neither on the contaminated Illinois River nor on the Mississippi, where the effluent from other nuclear plants upstream in Minnesota and Wisconsin might lead to rises in mortality.

And when we carried out the comparison in the change of infant mortality rates for these two groups of rural counties of similar climate, medical care, and socioeconomic character, the result was even more conclusive: While the mortality rates in the counties around the reactor had increased an average of 48 percent, the upwind control counties actually showed a decline of 2 percent in their average infant mortality rates.

Furthermore, with the prevailing westerly winds, the radioactive gas would drift eastward to Cook County, where Chicago was located, with a population of some five million. Since the radioactivity would have become much diluted with distance, only a small rise in mortality rates of a few percent would be likely. But since so many more children were born every year in Chicago than in Grundy County, the total number of additional deaths would be significant. And when we checked the figures, this is exactly what had taken place: Infant mortality in Cook County had gone up by 1.5 percent, during a time when in New York City it had declined by 6.7 percent.

Since some six million people lived within a radius of 50 to 60 miles from the Dresden reactor, and since the total population of Illinois

was ten million, there should have been a significant rise in infant mortality for Illinois as a whole. And there was indeed—from an all-time low point of 23.9 in 1963 to a peak in 1966 of 25.6, in exact coincidence with the peak of gaseous emissions from the Dresden reactor. This was followed by a renewed decline in both recorded gaseous releases and infant mortality as the defective fuel-rods were replaced.

With the advice of Dr. Morris DeGroot, head of the Statistics Department, Carnegie-Mellon University, who had become interested in the problem, we applied further statistical tests. The results were always the same: A significant rise and decline in infant mortality in Illinois compared to all other neighboring states in the northern U.S., correlating directly with the rise and decline of radioactive emissions from the Dresden reactor. Relative to Ohio, a few hundred miles to the east, where the infant mortality rate had been the same as in Illinois before the reactor had been started up in 1960, the excess infant deaths in Illinois for the years 1960–1968 numbered close to 4000. And for each infant dying in the first year of life, it was well known that there were perhaps three to four that would live with serious genetic defects, crippling congenital malformations, and mental retardation, afflictions in many ways far worse than death in early infancy.

The largest numbers of deaths among the newborn infants were caused by asphyxia or respiratory distress, including hyaline membrane disease,

long known to be associated with immaturity, and also general immaturity and "crib-death." These were the very causes that had risen sharply all over the world during the period of nuclear testing and had only begun to decline again a few years after the test-ban treaty came into force. Yet here in Illinois, they were still increasing. And among the older infants, noninfectious respiratory disease deaths rose almost 90 percent, and bronchitis almost 50 percent, in the two years after 1964.

In fact, for all ages, there was a rapid rise in deaths due to such lung diseases as emphysema and bronchitis after the onset of the Dresden emissions. The rise was far greater than in more heavily polluted New York. In the ten years between 1949 and 1959, these death rates in Illinois increased by only 9 percent, but they rose by 75 percent in the short period from 1959, when the reactor was completed, to 1966, the last year for which data was available. This was more than eight times the previous annual rate of increase.

Thus, the radioactive gases released from reactor stacks, gases which had been widely regarded as relatively harmless, now appeared to be far more serious in their effects than had been anticipated. Although these gases do not concentrate and remain in the human body, they do dissolve readily in the bloodstream and especially in the fatty parts of many cell membranes when they are inhaled over periods of hours or days.

And some of them transform themselves into the biologically damaging cesium, strontium, and yttrium inside the body. As a result, the internal radiation damage to the small air sacs of the lungs, which are lined with cells that produce a crucial fatty substance (lipid) that acts to keep these air sacs open when the air is exhaled, could be far more serious in causing respiratory damage than the external radiation dose from the radioactive gases outside the body.

There was still another way, more indirect but more efficient, in which small amounts of radioactivity could produce deaths from respiratory problems, especially in the newborn. Some of the radioactive chemicals produced by the fission of uranium—such as yttrium-90, the daughter product of strontium—were known to concentrate in the pituitary gland. And recent studies had revealed that the critical lipid needed to prevent the lung from collapsing was produced in special cells of the lung under the chemical control of the pituitary gland in the last few weeks of fetal development just before birth. Thus, even slight damage to the pituitary gland from radioactivity in the air or in the mother's diet could lead to a slight retardation in development, so that the lung would not be quite ready to function properly immediately after birth. And the result would be that otherwise apparently normal babies would be born underweight and would succumb to respiratory failure shortly after birth.

The rise in infant deaths from respiratory

diseases associated with immaturity also indicated that the atmospheric reactor releases should be causing an increase in low-birthweight babies. This expectation was confirmed by the data for Grundy County, where the Dresden reactor was located. The number of low-birthweight babies born in this county rose and declined in exact synchronism with the measured gaseous emissions, the rises going as high as 140 percent. No such increases in the number of underweight babies took place in the six control counties more than 40 miles west of the reactor.

The sudden rise in emphysema and bronchitis all over the United States and other countries, noted by I. M. Moriyama, following the onset of large-scale atmospheric releases of radioactive gas and dust in the early 1950s, also fitted the hypothesis that radioactivity in the air was causing lung damage. When we plotted the emphysema and bronchitis death rates for the states where ordinary air pollution was lowest but radioactivity in the air itself was highest, such as dry, dusty Wyoming, Utah, and New Mexico, where the winds picked up the radioactive dust again and again, we found that after declining in the 1940s, the respiratory death rates per 100,000 people suddenly began to rise sharply between 1946 and 1951, exceeding those in the much more polluted but higher-rainfall states of the east such as New York and Massachusetts, where the radioactivity was cleansed from the atmosphere and soaked into the ground by the rains.

By the early 1960s, even in the heavily polluted coal- and steel-producing state of Pennsylvania, these types of respiratory deaths, normally attributed only to ordinary air pollution, were lower than in the clean mountain air of Wyoming.

Clearly, air pollution from ordinary fossil-fuel-burning power plants, which had doubled steadily every ten years for many decades, could therefore not be blamed for all of the alarming rise in lung disease deaths. Instead, all the evidence pointed to radioactive air pollution, both from fallout and from nuclear power plants, as the greatest single contributor to the rise in all types of chronic lung disease around the world, multiplying the effects of the other pollutants—including cigarettes, as in the case of the uranium miners.

Furthermore, there was one source of radioactive pollution that was potentially even more serious than the boiling-water reactors. This was the effluent from the nuclear fuel-processing plants. These plants recovered uranium from the spent reactor fuel elements, as well as plutonium, which could be sold back to the government for use in building bombs and missile warheads. In the process, radioactive gases and large amounts of other fission products were discharged into the air and adjacent rivers. Here all the efforts to prevent the escape of radioactivity from the reactors themselves were therefore completely nullified.

The students gathered the data for the first

commercial fuel-reprocessing plant, located in West Valley, New York, some 25 miles south of Buffalo. This plant had gone into operation in April 1966, and reports on its radioactive gas releases, as well as measurements of liquid waste releases and doses from the food produced nearby, had just been published by the U.S. Bureau of Radiological Health in May 1970. When we looked at the available data for infant mortality in Cattaraugus County, where the plant was located, we saw that infant mortality had jumped up 54 percent between 1966 and 1967, far above the rate for New York State as a whole.

Once again, every adjacent county had gone up dramatically at the same time, while the next ring of counties rose only slightly. And those beyond 50 miles all showed declines in their infant death rate, as did New York State as a whole and all the adjoining New England states to the east.

But this situation was not confined to New York State. In some of the counties in Pennsylvania just to the south of the plant, the same rise in infant mortality had taken place. Warren County, directly to the southwest along the valley of the Allegheny River, had gone up almost exactly the same amount as Cattaraugus County. And along the Allegheny River below Cattaraugus County where the plant was located, infant mortality had either jumped up or refused to decline further, the effect diminishing with distance all the way down to Pittsburgh.

The map of the area made the explanation evident: The small tributaries that flowed into the Allegheny River as it passed through New York State originated within a few miles of the fuel-processing plant near West Valley, where the radioactivity discharged from the stack and the storage reservoirs seeped into the watershed for the entire Allegheny River system. Along the Allegheny near the Pennsylvania border, infant mortality had risen 56 percent in Warren County and 48 percent in Venango, through which it passed next. Even as far away as Armstrong County, more than 100 miles downriver, infant mortality had gone up 4 percent that same year, while Pennsylvania as a whole showed no such rise, though it was not declining as rapidly as rural states having no nuclear facilities. Evidently it was not just the inhalation of the atmospheric gases that was important in infant mortality, but also their deposition by rainfall in the headwaters of the Allegheny River, contaminating water, fish, milk, and vegetables with radioactive cesium, strontium, iodine, and other toxic elements.

When we checked the levels of radioactivity in milk reported by the Public Health Service, we found the confirmation of what we had begun to suspect: Of all the milk-sampling stations in the entire United States reporting for the 12-month period ending in March 1970, only those in Pennsylvania showed a level of short-lived iodine-131 greater than 1 micromicrocurie per liter. And there was a more disturbing

piece of evidence: Strontium-90 had climbed back up to more than half the level that existed at the peak of atmospheric weapons testing, and it was still rising. The West Valley plant was the first commercial facility of its type in the United States, and it emitted far greater amounts of toxic radioactive elements into the environment than did any single nuclear reactor, including Dresden. Yet many more of these plants were planned for the future.

There simply could be no further doubt as to the cause of the rising infant mortality around the West Valley plant: Measurements carried out by the Public Health Service and published in May 1970 showed that, aside from the dose produced by the krypton gases released into the air, doses as high as 250 millirads from cesium and 532 millirads from strontium would be received in a single year by any individuals who ate significant amounts of the area's heavily contaminated fish and deer. These were doses much greater than the 100 millirads normally received from natural background radiation. In fact, they far exceeded even the annual doses during the height of nuclear testing. And these dose calculations were only for the adult, and not for the much more sensitive fetus and infant, where the even more intensive concentration in various critical organs would make the doses far higher still.

The findings were further confirmed when the infant mortality rates in the counties around the AEC's Hanford Laboratories in Washington were

graphed. It was at Hanford that nuclear fuel was first processed to produce the plutonium for the Trinity explosion at Alamogordo, New Mexico, in 1945. During this period large quantities of radioactivity were released by the Hanford plant. These releases had explained the early infant mortality rise in Montana and North Dakota which showed up on the Trinity map. But at the time, I did not examine the effect on the counties around the Hanford works itself. Now, when we compared the infant mortality rate for 1945, after the emissions had occurred, with the rate for 1943, before the plant had been started up, we found that the rate for Benton County, where the plant was located, had jumped 160 percent. Umatilla, the adjoining county to the south, had gone up 60 percent, while Franklin, directly to the east, increased 50 percent and Walla Walla, just to the southeast of Franklin, rose 10 percent. Yet infant mortality for the state of Washington as a whole declined, as it also did in Oregon.

And within a few more months, the results for Consolidated Edison's Indian Point plant on the Hudson River in Westchester County, 20 miles north of New York City, disclosed similar rises and declines in infant mortality that correlated with the rises and falls in the plant's radioactive releases, including observable effects on New York City itself. Yet this was a pressurized-water reactor, the type which generally had the lowest releases of all, and it was located in an area of excellent medical care. Evidently it had

not been possible to maintain the standards of a naval-type plant and remain commercially competitive with the boiling-water plants. Similar situations existed around all the reactors we checked in various parts of the country. Even the small research-type reactors, such as the TRIGA, installed on college campuses and in laboratories all over the world, appeared to be capable of causing the same effect. When figures on the year-by-year emissions of the TRIGA reactor at Pennsylvania State College became available, we compared the infant mortality rates in the surrounding town, State College, with those in Lebanon City, a similar town some 100 miles to the east. State College showed precipitous rises and falls in infant mortality, corresponding closely with the rises and falls in emissions from the TRIGA. The State College rate went from 9.9 per 1000 births in 1963 to 24.7 in 1968. During the same period, the rate in Lebanon City, as well as in Pennsylvania as a whole, declined steadily from the peak reached during the atmospheric tests of 1961-62.

Since the population of State College was comparatively small, however, a remote possibility existed that these increases could be due to chance fluctuations. So we next examined infant mortality rates around the TRIGA on the University of Illinois campus in Urbana, where the population was much larger. From 1962, when the reactor commenced operation, through 1965, the year it reached full power, infant mortality

increased by 300 percent. In this study, for the first time, we also had an opportunity to look at another category of possible radiation effects: deaths from congenital malformations. During the same period in Urbana these deaths increased by 600 percent, from 3.5 per 100,000 in 1962 to 23.5 in 1965. And in 1968, after the reactor was shut off, they turned downward to 6.6 per 100,000, while infant mortality showed a similar drop. In McLean County, which extended 20 to 60 miles northwest of Urbana and thus would not have been significantly exposed to the effluent, both categories of death declined steadily throughout the same period. The surprising strength of the effects from the TRIGA emissions, which were much lower than the emissions from the larger reactors we studied, could be explained by the fact that the TRIGAS were located right in the middle of densely populated areas. Therefore, the emissions would reach the developing infants in much more concentrated form, with much less time for the short-lived isotopes to lose their radioactivity.

It was the announced intention of the AEC, numerous public utilities, and the government that this country's energy needs would be supplied largely by nuclear power reactors in the near future. Only fifteen or twenty such reactors were in actual operation, but more than a hundred were under construction or planned, as were the necessary number of fuel-reprocessing plants. But if our findings proved correct, then

the entire program, with its phenomenally large investment of funds and scientific energy, would become virtually useless in its present form. Considering the apparent effects from normal operation of these plants, during which no more than one ten-millionth of their stored-up radioactivity had ever been discharged, a single large accidental release could be a national catastrophe of nuclear warfare dimensions. If the general public grasped this fact, then most people would probably consider the risk of this technology far too great to be accepted. But through all the years while reactor technology was being developed, the possible dangers of low-level radiation—either from fallout or from nuclear power plants—had been publicly minimized by the military, by industry, and by the health agencies who had given their stamp of approval to nuclear activities. The warning signs had been ignored or suppressed. And little or no funds had been made available for development of the potentially safer and more efficient alternatives to nuclear power, such as coal gassification or Magnetohydrodynamics, which would permit the continued use of the still-enormous reserves of fossil fuels. Little or nothing was done to find means of harnessing the vast stores of geothermal energy in the crust of the earth, or the pollution-free energy of the sun. Yet there was little question that these alternative means of electric power production could have been successfully developed.

Our reactor findings were met with opposition

as strenuous as that which greeted the evidence on the effects of fallout. Notable among our critics was Edythdalena Tompkins, a public health scientist who was recently placed in charge of all studies of radiation effects on the population by the U.S. Environmental Protection Agency. Edythdalena Tompkins was also the wife of Paul C. Tompkins, the director of the Federal Radiation Council who in 1964 had presided over the first raising of permissible radiation levels in history. Additionally, she had been a critic of the fallout evidence, particularly that relating to the effects of the Trinity test. In the spring of 1970, a student at the Pittsburgh School of Public Health had informed me that Mrs. Tompkins had told him there were serious errors in my map of infant mortality after the Trinity explosion. My map had shown no rise in infant mortality among the white population in three states that were in the path of the fallout—Oklahoma, Florida, and North Carolina. The explanation for this was that according to the official weather map these states had received little or no rain and thus little fallout during the week following the Trinity explosion. This fact had provided an important confirmation of my hypothesis. However, Mrs. Tompkins had told this student that just the opposite had been the case: the infant mortality in these states had actually *risen*, just as it had in the states that received the rains. He then gave me a series of five maps which had been prepared by Mrs. Tompkins, and on all of these maps the three key

states did indeed show sharp increases in white infant mortality during the five years following Trinity.

This appeared to be a devastating piece of evidence, but my associates and I rechecked our figures and found that only in these three crucial states did they differ from those on Mrs. Tompkins's maps. I suggested to the student that he himself recalculate the figures. After doing so he informed me that our figures were the correct ones and then called Mrs. Tompkins for an explanation. Mrs. Tompkins said that she had evidently made a mistake, but that in any event these maps were not intended for publication. Subsequently, however, AEC representatives and members of the Joint Committee on Atomic Energy stated publicly that my infant mortality figures for the states that had low rainfalls after the Trinity test had been proven to be inaccurate and that the true figures completely invalidated my conclusions. Yet the only time these particular figures had ever been challenged was by Mrs. Tompkins.

After our group began making the reactor findings public, Mrs. Tompkins, by now with the newly formed Environmental Protection Agency to which she and her husband had been transferred, began conducting her own studies of this subject too. Her method was to calculate the infant mortality rates in a series of circular areas surrounding the reactor, and compare these figures for five-year periods before and after the

reactor had gone into operation. She concluded that even the most heavily emitting boiling-water reactors had no detectable effect on infant mortality.

Since vital statistics are recorded by county and not by circular regions around reactors, however, Mrs. Tompkins's method first of all necessitated that she make her own estimates of both the population figures and the infant mortality rates. And her use of concentric ring-shaped areas omitted a very important consideration: The emissions from reactors are not evenly distributed in all directions. Their distribution depends not only on the direction of the prevailing winds, or on geographical features such as high mountains and resulting differences in rainfall, but also on the discretion of the reactor engineers, who can and do time the releases to coincide with a certain wind direction which may or may not be the prevalent one. And then, of course, the counties which take their drinking water and fish from the rivers, lakes, or oceans into which the reactor releases its liquid effluent would also be expected to show sharper increases than the others. Thus, highly asymmetrical situations can develop around reactors, situations in which the counties most heavily exposed to the effluent show sharp rises and falls in infant mortality which correlate directly with rises and falls in the reactor releases, while in other counties, such as those upwind to the west of Dresden, the rate may continue the decline that began shortly after the cessation of

atmospheric testing. Thus, if all the surrounding counties are averaged together over five-year periods as in Mrs. Tompkins's method, the overall figure may show little or no increase in infant mortality. Furthermore, in the case of reactors which began operation in the early 1960s in areas which had received heavy fallout (as had the three boiling-water reactors studied by Mrs. Tompkins), it is possible by this technique to demonstrate an actual *decline* in infant mortality after the reactors were started up and the fallout levels died down. But, in all of these cases, if one examines the yearly figures, the infant mortality rates in the counties heavily exposed to the reactor effluent show sharp rises and falls in direct correlation with the releases, declining steadily with distance in any direction from the reactor when the counties are of similar socioeconomic and climatic character.

Significantly, an independent statistical study of this subject was presented at a scientific meeting in July 1971 by Dr. Morris H. DeGroot, head of the Department of Mathematical Statistics at Carnegie-Mellon University. Dr. DeGroot found that infant mortality increases did take place in close correlation with releases of radioactivity from the heavily emitting reactors at Dresden, Illinois; Indian Point, New York; and Brookhaven, Long Island. Perhaps most important was his finding that in the area around the reactor at Shippingport, Pennsylvania—the only other

reactor studied by Dr. DeGroot—there was no correlation between releases and changes in infant mortality. As the official release figures showed, the Shippingport reactor had the lowest gaseous emissions of any reactor in the country, since it was a non-commercial naval submarine type of plant.

But later in 1971, the most comprehensive independent study of all was completed. It was conducted by Dr. Lester B. Lave and his associates, Dr. Samuel Leinhardt and Martin B. Kaye, of the Graduate School of Business Administration at Carnegie-Mellon University. This was a study of fallout effects, but the results apply equally to reactor emissions. The three scientists concluded that, during the time period studied (1961–67), fallout appears to have been the single most important factor affecting fetal, infant, and adult mortality, more important than ordinary air pollution. Through the use of computerized statistical techniques they corrected their estimates to account for the effects of such variables as sulfur dioxide, socioeconomic factors, background radiation, and others in 61 metropolitan areas of the United States. The principal findings and their implications may be briefly summarized as follows:

Infant mortality is strongly associated with levels of strontium-90 and cesium-137 in milk, especially the former. The association is such that for every single micromicrocurie of strontium-90

per liter of milk there is an increase of 12 infant deaths per 100,000 births. Since, during 1961–67, there was an average of 15.8 micromicrocuries per liter of milk in the U.S., then these findings indicate that during this period there were close to 7600 infant deaths *every year* due to fallout. For the world population, this would mean an extra 100,000 infant deaths per year. But during the peak of testing, these levels reached between 50 and 100 micromicrocuries per liter in many locations around the world, and today they are still between 5 and 15 in most parts of the northern hemisphere. And they are in fact *now rising again* following the large French and Chinese test series and the rapid growth in releases from nuclear reactors and fuel reprocessing plants.

Dr. Lave's group also found that mortality rates for the whole population—in other words, all causes of death among all ages—were also highly correlated with fallout levels. The calculations showed that there were 1.29 extra deaths per 100,000 people for each single micromicrocurie of strontium-90 per liter of milk. At the 1961–1967 levels, this amounts to some 40,000 extra deaths each year in the United States, and thus some 600,000 among the world's population of over three billion people.

And during the fifteen-year period of heavy nuclear testing that began in the early 1950s, when the short-lived iodine and other isotopes

were added to the strontium-90 in the milk, there would have been many millions of extra deaths.

Now, more than a quarter of a century after Hiroshima, the studies are being made.

Epilogue

On December 18, 1970, more than seventeen years after Professor Clark and his students accidentally discovered the rainout in Troy, New York, the U.S. Atomic Energy Commission conducted an underground nuclear weapons test at the Nevada Test Site. Code-named "Baneberry," this Hiroshima-sized bomb was exploded some 800 feet underground. The explosion opened up a fissure in the rock and large quantities of radioactivity escaped upward into the atmosphere. In the vicinity of the test site, the slowly-drifting cloud of radioactive dust produced readings of 25 rads per hour on the ground. Hundreds of employees were seriously overexposed and had to be quickly evacuated.

Shortly after the accident, the AEC's Division of Biology and Medicine and the Utah State Division of Health notified a team of fallout specialists working at the University of Utah under contract to the AEC. The team, headed by Drs. Robert C. Pendleton and Charles W. Mays, scientists who had long warned of the dangers to the infant thyroid from radioactive iodine, immediately set about determining the direction and

intensity of the radioactive clouds. From the Salt Lake Weather Bureau they learned the speed of the winds at the time of detonation, while the Nevada Operations Office informed them that the radioactive leakage had occurred during a period of very strong wind shear, with winds at different altitudes blowing in different directions. The team was able to estimate that during the next twenty-four hours the lowest part of the cloud would probably go to the east of Winnemucca, Nevada, while the layer above would be blown across Utah to the southeast. The next-highest layer was apparently headed for New Mexico, while the topmost parts were expected to be carried into Utah between Highways 56 and 21.

Once the direction of the fallout was estimated, the extensive network of sampling stations around the state, constructed in the years since the dangers from fallout in the milk were discovered, could go into operation, estimating the strength of the radioactivity in the air and on the ground. This was indeed a far more sophisticated operation than the one mounted so long before by Dr. Clark's students, driving from town to town in their jalopies carrying rudimentary Geiger counters. And in this case, there was no problem about estimating the internal radiation dose, something which had not even been considered in Troy back in 1953. Now, a program run jointly with the Utah Division of Fish and Game was put into action to procure samples of

the local wildlife which had been in the path of the fallout. Conservation officers were enlisted to collect deer that had recently been killed on the highways, while sheep, deer, and rabbits were shot in areas where they were regarded as a nuisance. These animals were to be dissected in the laboratory, where the concentrations of isotopes in their body organs would be measured.

On the nineteenth and twentieth of December, sampling teams departed from Salt Lake City to obtain samples of snow and vegetation. Instead of the burdock leaves favored by Dr. Clark and his students, the fallout specialists took samples of alfalfa, sagebrush, and juniper. Some of the alfalfa samples consisted of loose hay from exposed bales.

According to the laboratory results, the most prominent gamma-radiation emitting isotopes in the Baneberry fallout were the short-lived, intensely radioactive iodine-131 and iodine-133. These were found in the lungs, thyroids, stomachs, and fetuses of deer and sheep, as well as in snow, milk, and vegetation. The fallout specialists were able to determine that if the Baneberry explosion had happened to take place in the warmer months, when the cows were out to pasture, the total dose to the local children from iodine in the milk would have been approximately 120 millirads, with some receiving much higher exposures. This did not include the dose from the other isotopes inhaled or eaten. Furthermore, the scientists observed that low wind velocities

and an atmospheric inversion had fortunately served to keep the fallout that reached Utah fairly stationary for a number of days in a position over relatively unpopulated areas. This allowed quantities of the heaviest fallout particles to settle to the ground, while the short-lived isotopes lost much of their radioactivity before the cloud was blown over the more populated areas. If these fortuitous circumstances had not existed, the doses might have been much higher for the people of Utah, perhaps as high as in Troy of 1953.

As it turned out, the predictions of where the fallout would drift were wrong. The heaviest clouds went north and northeast toward Idaho, Washington, and Montana, where rain and snow brought down much more radioactivity than in Utah. Subsequently, the fallout from Baneberry was detected across the northern U.S. by large rises in the cesium-137 levels in milk, as could be seen in the state-by-state tabulation of cesium levels for December 1970 printed in the April 1971 issue of *Radiation Health Data and Reports*, a U.S. Public Health Service publication. Data for the radioactivity on the ground also showed that the fallout had drifted into Canada, thus violating the provisions of the 1963 Test-Ban Treaty, which does not permit nuclear tests that release radioactivity beyond the national borders of the nation conducting them.

But as far as the general public was concerned, there was only the following statement by the

AEC, carried by *The New York Times* and the rest of the press around the country:

The Commission said that the radioactivity was of such low intensity that it presented no danger. It was detected at altitudes of several thousand feet, and only the most minute traces of radioactive contamination would reach the ground, the Commission said. The AEC . . . said that nowhere outside the immediate area of the test was the fallout dangerous to human life or health.

In the spring of 1971 our group gathered the data for radioactivity in the air, milk, and on the ground both before and after the Baneberry test. This was then correlated with the mortality figures for infants born following the explosion, as reported in the U.S. Monthly Vital Statistics. In all of the states where the total radioactivity rose highest—Idaho, Montana, Oregon, Nevada, Washington, Nebraska, and as far away as Minnesota and Maine—infant mortality also rose sharply during the first three months after the test. Across the rest of the U.S., the pattern of general decline continued.

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Nuclear bomb tests and nuclear reactors have raised radiation levels in the United States and elsewhere in the world. Government agencies have assured us over the years that safe radiation levels have not been exceeded.

However, for over ten years, Dr. Ernest J. Sternglass, professor of radiation physics at the University of Pittsburgh, has waged a battle to bring the evidence of the harmful effects of low-level radiation before the scientific community and the world.

Here is his story and his evidence—decide for yourself.

